

```
#Q1. Import the csv data to Dataframe
```

```
import numpy as np
```

```
import pandas as pd
```

```
df =
```

```
pd.read_csv('C:/Users/Admin/OneDrive/Desktop/IoT_Academy/Assignments/Tests/netflix_titles.csv')
```

```
df
```

|      | show_id | type    | title                 | director        | \   |
|------|---------|---------|-----------------------|-----------------|-----|
| 0    | s1      | Movie   | Dick Johnson Is Dead  | Kirsten Johnson |     |
| 1    | s2      | TV Show | Blood & Water         | NaN             |     |
| 2    | s3      | TV Show | Ganglands             | Julien Leclercq |     |
| 3    | s4      | TV Show | Jailbirds New Orleans | NaN             |     |
| 4    | s5      | TV Show | Kota Factory          | NaN             |     |
| ...  | ...     | ...     | ...                   | ...             | ... |
| 8775 | s8803   | Movie   | Zodiac                | David Fincher   |     |
| 8776 | s8804   | TV Show | Zombie Dumb           | NaN             |     |
| 8777 | s8805   | Movie   | Zombieland            | Ruben Fleischer |     |
| 8778 | s8806   | Movie   | Zoom                  | Peter Hewitt    |     |
| 8779 | s8807   | Movie   | Zubaan                | Mozez Singh     |     |

|      | cast                                              | country       |
|------|---------------------------------------------------|---------------|
| \    |                                                   |               |
| 0    | NaN                                               | United States |
| 1    | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... | South Africa  |
| 2    | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... | NaN           |
| 3    | NaN                                               | NaN           |
| 4    | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... | India         |
| ...  | ...                                               | ...           |
| 8775 | Mark Ruffalo, Jake Gyllenhaal, Robert Downey J... | United States |
| 8776 | NaN                                               | NaN           |
| 8777 | Jesse Eisenberg, Woody Harrelson, Emma Stone, ... | United States |
| 8778 | Tim Allen, Courteney Cox, Chevy Chase, Kate Ma... | United States |
| 8779 | Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan... | India         |

|   | date_added         | release_year | rating | duration  | \ |
|---|--------------------|--------------|--------|-----------|---|
| 0 | September 25, 2021 | 2020         | PG-13  | 90 min    |   |
| 1 | September 24, 2021 | 2021         | TV-MA  | 2 Seasons |   |
| 2 | September 24, 2021 | 2021         | TV-MA  | 1 Season  |   |
| 3 | September 24, 2021 | 2021         | TV-MA  | 1 Season  |   |

|      |                    |      |       |           |
|------|--------------------|------|-------|-----------|
| 4    | September 24, 2021 | 2021 | TV-MA | 2 Seasons |
| ...  |                    | ...  | ...   | ...       |
| 8775 | November 20, 2019  | 2007 | R     | 158 min   |
| 8776 | July 1, 2019       | 2018 | TV-Y7 | 2 Seasons |
| 8777 | November 1, 2019   | 2009 | R     | 88 min    |
| 8778 | January 11, 2020   | 2006 | PG    | 88 min    |
| 8779 | March 2, 2019      | 2015 | TV-14 | 111 min   |

|      |                                                   |
|------|---------------------------------------------------|
|      | listed_in                                         |
| 0    | Documentaries                                     |
| 1    | International TV Shows, TV Dramas, TV Mysteries   |
| 2    | Crime TV Shows, International TV Shows, TV Act... |
| 3    | Docuseries, Reality TV                            |
| 4    | International TV Shows, Romantic TV Shows, TV ... |
| ...  | ...                                               |
| 8775 | Cult Movies, Dramas, Thrillers                    |
| 8776 | Kids' TV, Korean TV Shows, TV Comedies            |
| 8777 | Comedies, Horror Movies                           |
| 8778 | Children & Family Movies, Comedies                |
| 8779 | Dramas, International Movies, Music & Musicals    |

[8780 rows x 11 columns]

*#a) to display first 5 records in dataframe*  
df.head()

|   | show_id | type    | title                 | director        | \ |
|---|---------|---------|-----------------------|-----------------|---|
| 0 | s1      | Movie   | Dick Johnson Is Dead  | Kirsten Johnson |   |
| 1 | s2      | TV Show | Blood & Water         | NaN             |   |
| 2 | s3      | TV Show | Ganglands             | Julien Leclercq |   |
| 3 | s4      | TV Show | Jailbirds New Orleans | NaN             |   |
| 4 | s5      | TV Show | Kota Factory          | NaN             |   |

|   | cast                                              | country       | \ |
|---|---------------------------------------------------|---------------|---|
| 0 | NaN                                               | United States |   |
| 1 | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... | South Africa  |   |
| 2 | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... | NaN           |   |
| 3 | NaN                                               | NaN           |   |
| 4 | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... | India         |   |

|   | date_added         | release_year | rating | duration  | \ |
|---|--------------------|--------------|--------|-----------|---|
| 0 | September 25, 2021 | 2020         | PG-13  | 90 min    |   |
| 1 | September 24, 2021 | 2021         | TV-MA  | 2 Seasons |   |
| 2 | September 24, 2021 | 2021         | TV-MA  | 1 Season  |   |
| 3 | September 24, 2021 | 2021         | TV-MA  | 1 Season  |   |
| 4 | September 24, 2021 | 2021         | TV-MA  | 2 Seasons |   |

|   |                                                 |
|---|-------------------------------------------------|
|   | listed_in                                       |
| 0 | Documentaries                                   |
| 1 | International TV Shows, TV Dramas, TV Mysteries |

```

2 Crime TV Shows, International TV Shows, TV Act...
3 Docuseries, Reality TV
4 International TV Shows, Romantic TV Shows, TV ...

```

*#a) to display last 5 records in dataframe*

```
df.tail()
```

|      | show_id | type    | title       | director \      |
|------|---------|---------|-------------|-----------------|
| 8775 | s8803   | Movie   | Zodiac      | David Fincher   |
| 8776 | s8804   | TV Show | Zombie Dumb | NaN             |
| 8777 | s8805   | Movie   | Zombieland  | Ruben Fleischer |
| 8778 | s8806   | Movie   | Zoom        | Peter Hewitt    |
| 8779 | s8807   | Movie   | Zubaan      | Mozez Singh     |

|      | cast                                              | country       |
|------|---------------------------------------------------|---------------|
| 8775 | Mark Ruffalo, Jake Gyllenhaal, Robert Downey J... | United States |
| 8776 | NaN                                               | NaN           |
| 8777 | Jesse Eisenberg, Woody Harrelson, Emma Stone, ... | United States |
| 8778 | Tim Allen, Courteney Cox, Chevy Chase, Kate Ma... | United States |
| 8779 | Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan... | India         |

|      | date_added        | release_year | rating | duration \ |
|------|-------------------|--------------|--------|------------|
| 8775 | November 20, 2019 | 2007         | R      | 158 min    |
| 8776 | July 1, 2019      | 2018         | TV-Y7  | 2 Seasons  |
| 8777 | November 1, 2019  | 2009         | R      | 88 min     |
| 8778 | January 11, 2020  | 2006         | PG     | 88 min     |
| 8779 | March 2, 2019     | 2015         | TV-14  | 111 min    |

|      | listed_in                                      |
|------|------------------------------------------------|
| 8775 | Cult Movies, Dramas, Thrillers                 |
| 8776 | Kids' TV, Korean TV Shows, TV Comedies         |
| 8777 | Comedies, Horror Movies                        |
| 8778 | Children & Family Movies, Comedies             |
| 8779 | Dramas, International Movies, Music & Musicals |

*#b) convert dataframe to array*

```
array = df.to_numpy()
```

*#b) getting array shape (rows,columns)*

```
array.shape
```

```
(8780, 11)
```

*#c) show all column names in a dataframe*

```
print(list(df.columns))
```

```
['show_id', 'type', 'title', 'director', 'cast', 'country',
'date_added', 'release_year', 'rating', 'duration', 'listed_in']
```

*#d) to get descriptive statistics of all variables in dataframe*  
df.describe(include='all')

|        | show_id | type  | title                | director \    |
|--------|---------|-------|----------------------|---------------|
| count  | 8780    | 8780  | 8780                 | 6150          |
| unique | 8780    | 2     | 8780                 | 4516          |
| top    | s1      | Movie | Dick Johnson Is Dead | Rajiv Chilaka |
| freq   | 1       | 6110  | 1                    | 19            |
| mean   | NaN     | NaN   | NaN                  | NaN           |
| std    | NaN     | NaN   | NaN                  | NaN           |
| min    | NaN     | NaN   | NaN                  | NaN           |
| 25%    | NaN     | NaN   | NaN                  | NaN           |
| 50%    | NaN     | NaN   | NaN                  | NaN           |
| 75%    | NaN     | NaN   | NaN                  | NaN           |
| max    | NaN     | NaN   | NaN                  | NaN           |

|                | cast               | country       | date_added      |
|----------------|--------------------|---------------|-----------------|
| release_year \ |                    |               |                 |
| count          | 7956               | 7952          | 8770            |
| 8780.000000    |                    |               |                 |
| unique         | 7671               | 745           | 1765            |
| NaN            |                    |               |                 |
| top            | David Attenborough | United States | January 1, 2020 |
| NaN            |                    |               |                 |
| freq           | 19                 | 2809          | 108             |
| NaN            |                    |               |                 |
| mean           | NaN                | NaN           | NaN             |
| 2014.178474    |                    |               |                 |
| std            | NaN                | NaN           | NaN             |
| 8.827938       |                    |               |                 |
| min            | NaN                | NaN           | NaN             |
| 1925.000000    |                    |               |                 |
| 25%            | NaN                | NaN           | NaN             |
| 2013.000000    |                    |               |                 |
| 50%            | NaN                | NaN           | NaN             |
| 2017.000000    |                    |               |                 |
| 75%            | NaN                | NaN           | NaN             |
| 2019.000000    |                    |               |                 |
| max            | NaN                | NaN           | NaN             |
| 2021.000000    |                    |               |                 |

|        | rating | duration | listed_in                    |
|--------|--------|----------|------------------------------|
| count  | 8776   | 8780     | 8780                         |
| unique | 14     | 220      | 514                          |
| top    | TV-MA  | 1 Season | Dramas, International Movies |
| freq   | 3198   | 1788     | 361                          |
| mean   | NaN    | NaN      | NaN                          |

|     |     |     |     |
|-----|-----|-----|-----|
| std | NaN | NaN | NaN |
| min | NaN | NaN | NaN |
| 25% | NaN | NaN | NaN |
| 50% | NaN | NaN | NaN |
| 75% | NaN | NaN | NaN |
| max | NaN | NaN | NaN |

*#e) check the no.of unique values for each column*

```
df.nunique()
```

|              |      |
|--------------|------|
| show_id      | 8780 |
| type         | 2    |
| title        | 8780 |
| director     | 4516 |
| cast         | 7671 |
| country      | 745  |
| date_added   | 1765 |
| release_year | 74   |
| rating       | 14   |
| duration     | 220  |
| listed_in    | 514  |

dtype: int64

*#f) check the % of missing values for each column*

```
print(df.isnull().mean() * 100)
```

|              |           |
|--------------|-----------|
| show_id      | 0.000000  |
| type         | 0.000000  |
| title        | 0.000000  |
| director     | 29.954442 |
| cast         | 9.384966  |
| country      | 9.430524  |
| date_added   | 0.113895  |
| release_year | 0.000000  |
| rating       | 0.045558  |
| duration     | 0.000000  |
| listed_in    | 0.000000  |

dtype: float64

*#g) delete all rows where director column has missing values*

```
df_g=df.copy()
```

```
df_g.dropna(subset=['director'])
```

|      | show_id | type    | title \                          |
|------|---------|---------|----------------------------------|
| 0    | s1      | Movie   | Dick Johnson Is Dead             |
| 2    | s3      | TV Show | Ganglands                        |
| 5    | s6      | TV Show | Midnight Mass                    |
| 6    | s7      | Movie   | My Little Pony: A New Generation |
| 7    | s8      | Movie   | Sankofa                          |
| ...  | ...     | ...     | ...                              |
| 8774 | s8802   | Movie   | Zinzana                          |

|      |       |       |            |
|------|-------|-------|------------|
| 8775 | s8803 | Movie | Zodiac     |
| 8777 | s8805 | Movie | Zombieland |
| 8778 | s8806 | Movie | Zoom       |
| 8779 | s8807 | Movie | Zubaan     |

|      | director \                    |
|------|-------------------------------|
| 0    | Kirsten Johnson               |
| 2    | Julien Leclercq               |
| 5    | Mike Flanagan                 |
| 6    | Robert Cullen, José Luis Ucha |
| 7    | Haile Gerima                  |
| ...  | ...                           |
| 8774 | Majid Al Ansari               |
| 8775 | David Fincher                 |
| 8777 | Ruben Fleischer               |
| 8778 | Peter Hewitt                  |
| 8779 | Mozez Singh                   |

|      | cast \                                            |
|------|---------------------------------------------------|
| 0    | NaN                                               |
| 2    | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... |
| 5    | Kate Siegel, Zach Gilford, Hamish Linklater, H... |
| 6    | Vanessa Hudgens, Kimiko Glenn, James Marsden, ... |
| 7    | Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D... |
| ...  | ...                                               |
| 8774 | Ali Suliman, Saleh Bakri, Yasa, Ali Al-Jabri, ... |
| 8775 | Mark Ruffalo, Jake Gyllenhaal, Robert Downey J... |
| 8777 | Jesse Eisenberg, Woody Harrelson, Emma Stone, ... |
| 8778 | Tim Allen, Courteney Cox, Chevy Chase, Kate Ma... |
| 8779 | Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan... |

|              | country                                                         |
|--------------|-----------------------------------------------------------------|
| date_added \ |                                                                 |
| 0            | United States September 25,                                     |
| 2021         |                                                                 |
| 2            | NaN September 24,                                               |
| 2021         |                                                                 |
| 5            | NaN September 24,                                               |
| 2021         |                                                                 |
| 6            | NaN September 24,                                               |
| 2021         |                                                                 |
| 7            | United States, Ghana, Burkina Faso, United Kin... September 24, |
| 2021         |                                                                 |
| ...          | ...                                                             |
| ...          |                                                                 |
| 8774         | United Arab Emirates, Jordan March 9,                           |
| 2016         |                                                                 |
| 8775         | United States November 20,                                      |
| 2019         |                                                                 |
| 8777         | United States November 1,                                       |

|      |               |             |
|------|---------------|-------------|
| 2019 |               |             |
| 8778 | United States | January 11, |
| 2020 |               |             |
| 8779 | India         | March 2,    |
| 2019 |               |             |

|      | release_year | rating | duration \ |
|------|--------------|--------|------------|
| 0    | 2020         | PG-13  | 90 min     |
| 2    | 2021         | TV-MA  | 1 Season   |
| 5    | 2021         | TV-MA  | 1 Season   |
| 6    | 2021         | PG     | 91 min     |
| 7    | 1993         | TV-MA  | 125 min    |
| ...  | ...          | ...    | ...        |
| 8774 | 2015         | TV-MA  | 96 min     |
| 8775 | 2007         | R      | 158 min    |
| 8777 | 2009         | R      | 88 min     |
| 8778 | 2006         | PG     | 88 min     |
| 8779 | 2015         | TV-14  | 111 min    |

|      | listed_in                                         |
|------|---------------------------------------------------|
| 0    | Documentaries                                     |
| 2    | Crime TV Shows, International TV Shows, TV Act... |
| 5    | TV Dramas, TV Horror, TV Mysteries                |
| 6    | Children & Family Movies                          |
| 7    | Dramas, Independent Movies, International Movies  |
| ...  | ...                                               |
| 8774 | Dramas, International Movies, Thrillers           |
| 8775 | Cult Movies, Dramas, Thrillers                    |
| 8777 | Comedies, Horror Movies                           |
| 8778 | Children & Family Movies, Comedies                |
| 8779 | Dramas, International Movies, Music & Musicals    |

[6150 rows x 11 columns]

*#h) records with matching string in a column value*

df\_h=df.copy()

df\_h[df\_h['country'].str.contains('Germany', na=False)]

|      | show_id | type  | title \                             |
|------|---------|-------|-------------------------------------|
| 7    | s8      | Movie | Sankofa                             |
| 12   | s13     | Movie | Je Suis Karl                        |
| 129  | s130    | Movie | An Unfinished Life                  |
| 142  | s143    | Movie | Freedom Writers                     |
| 172  | s173    | Movie | School of Rock                      |
| ...  | ...     | ...   | ...                                 |
| 8590 | s8618   | Movie | Trash                               |
| 8634 | s8662   | Movie | Unfinished Song                     |
| 8641 | s8669   | Movie | V for Vendetta                      |
| 8702 | s8730   | Movie | Where the Money Is                  |
| 8718 | s8746   | Movie | Willy Wonka & the Chocolate Factory |

|        | director                                                               |
|--------|------------------------------------------------------------------------|
| cast \ |                                                                        |
| 7      | Haile Gerima Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...         |
| 12     | Christian Schwochow Luna Wedler, Jannis Niewöhner, Milan Peschel, ...  |
| 129    | Lasse Hallström Robert Redford, Jennifer Lopez, Morgan Freeman...      |
| 142    | Richard LaGravenese Hilary Swank, Patrick Dempsey, Scott Glenn, Im...  |
| 172    | Richard Linklater Jack Black, Joan Cusack, Mike White, Sarah Sil...    |
| ...    | ...                                                                    |
| ...    |                                                                        |
| 8590   | Stephen Daldry Wagner Moura, Martin Sheen, Rooney Mara, Selto...       |
| 8634   | Paul Andrew Williams Terence Stamp, Gemma Arterton, Christopher Ecc... |
| 8641   | James McTeigue Natalie Portman, Hugo Weaving, Stephen Rea, St...       |
| 8702   | Marek Kaniévski Paul Newman, Linda Fiorentino, Dermot Mulroney...      |
| 8718   | Mel Stuart Gene Wilder, Jack Albertson, Peter Ostrum, Roy...           |

|              | country                                                              |
|--------------|----------------------------------------------------------------------|
| date_added \ |                                                                      |
| 7            | United States, Ghana, Burkina Faso, United Kin... September 24, 2021 |
| 12           | Germany, Czech Republic September 23, 2021                           |
| 129          | Germany, United States September 1, 2021                             |
| 142          | Germany, United States September 1, 2021                             |
| 172          | United States, Germany September 1, 2021                             |
| ...          | ...                                                                  |
| ...          |                                                                      |
| 8590         | United Kingdom, Brazil, Germany January 1, 2019                      |
| 8634         | United Kingdom, Germany July 22, 2019                                |
| 8641         | United States, United Kingdom, Germany October 1, 2018               |
| 8702         | Germany, United States, United Kingdom, Canada January 15, 2020      |
| 8718         | United States, East Germany, West Germany January 1,                 |



2020

|      | release_year | rating | duration | \ |
|------|--------------|--------|----------|---|
| 7    | 1993         | TV-MA  | 125 min  |   |
| 12   | 2021         | TV-MA  | 127 min  |   |
| 129  | 2005         | PG-13  | 108 min  |   |
| 142  | 2007         | PG-13  | 124 min  |   |
| 172  | 2003         | PG-13  | 110 min  |   |
| ...  | ...          | ...    | ...      |   |
| 8590 | 2014         | R      | 114 min  |   |
| 8634 | 2012         | PG-13  | 94 min   |   |
| 8641 | 2005         | R      | 132 min  |   |
| 8702 | 2000         | PG-13  | 89 min   |   |
| 8718 | 1971         | G      | 100 min  |   |

|      | listed_in                                         |
|------|---------------------------------------------------|
| 7    | Dramas, Independent Movies, International Movies  |
| 12   | Dramas, International Movies                      |
| 129  | Dramas                                            |
| 142  | Dramas                                            |
| 172  | Comedies, Music & Musicals                        |
| ...  | ...                                               |
| 8590 | Dramas, Independent Movies, Thrillers             |
| 8634 | Comedies, Dramas, Independent Movies              |
| 8641 | Action & Adventure, Dramas, Sci-Fi & Fantasy      |
| 8702 | Action & Adventure, Comedies, Dramas              |
| 8718 | Children & Family Movies, Classic Movies, Come... |

[231 rows x 11 columns]

*#i) split column into 2 separate cols based on separator*

```
dfi= df.copy()
split= dfi['duration'].str.split(' ', expand=True)
dfi[['duration_Num', 'duration_Str']] = split
dfi.drop('duration', axis=1,inplace=True)
dfi
```

|      | show_id | type    | title                 | director        | \ |
|------|---------|---------|-----------------------|-----------------|---|
| 0    | s1      | Movie   | Dick Johnson Is Dead  | Kirsten Johnson |   |
| 1    | s2      | TV Show | Blood & Water         | NaN             |   |
| 2    | s3      | TV Show | Ganglands             | Julien Leclercq |   |
| 3    | s4      | TV Show | Jailbirds New Orleans | NaN             |   |
| 4    | s5      | TV Show | Kota Factory          | NaN             |   |
| ...  | ...     | ...     | ...                   | ...             |   |
| 8775 | s8803   | Movie   | Zodiac                | David Fincher   |   |
| 8776 | s8804   | TV Show | Zombie Dumb           | NaN             |   |
| 8777 | s8805   | Movie   | Zombieland            | Ruben Fleischer |   |
| 8778 | s8806   | Movie   | Zoom                  | Peter Hewitt    |   |
| 8779 | s8807   | Movie   | Zubaan                | Mozez Singh     |   |

|      |                                                   | cast | country       |
|------|---------------------------------------------------|------|---------------|
| \    | 0                                                 | NaN  | United States |
| 1    | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... |      | South Africa  |
| 2    | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... |      | NaN           |
| 3    |                                                   |      | NaN           |
| 4    | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... |      | India         |
| ...  |                                                   | ...  | ...           |
| 8775 | Mark Ruffalo, Jake Gyllenhaal, Robert Downey J... |      | United States |
| 8776 |                                                   |      | NaN           |
| 8777 | Jesse Eisenberg, Woody Harrelson, Emma Stone, ... |      | United States |
| 8778 | Tim Allen, Courteney Cox, Chevy Chase, Kate Ma... |      | United States |
| 8779 | Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan... |      | India         |

  

|      |                    | date_added | release_year | rating | \ |
|------|--------------------|------------|--------------|--------|---|
| 0    | September 25, 2021 |            | 2020         | PG-13  |   |
| 1    | September 24, 2021 |            | 2021         | TV-MA  |   |
| 2    | September 24, 2021 |            | 2021         | TV-MA  |   |
| 3    | September 24, 2021 |            | 2021         | TV-MA  |   |
| 4    | September 24, 2021 |            | 2021         | TV-MA  |   |
| ...  |                    | ...        | ...          | ...    |   |
| 8775 | November 20, 2019  |            | 2007         | R      |   |
| 8776 | July 1, 2019       |            | 2018         | TV-Y7  |   |
| 8777 | November 1, 2019   |            | 2009         | R      |   |
| 8778 | January 11, 2020   |            | 2006         | PG     |   |
| 8779 | March 2, 2019      |            | 2015         | TV-14  |   |

  

|              |                                                   | listed_in |     |
|--------------|---------------------------------------------------|-----------|-----|
| duration_Num | \                                                 |           |     |
| 0            | Documentaries                                     |           | 90  |
| 1            | International TV Shows, TV Dramas, TV Mysteries   |           | 2   |
| 2            | Crime TV Shows, International TV Shows, TV Act... |           | 1   |
| 3            | Docuseries, Reality TV                            |           | 1   |
| 4            | International TV Shows, Romantic TV Shows, TV ... |           | 2   |
| ...          |                                                   | ...       | ... |

|      |                                                |     |
|------|------------------------------------------------|-----|
| 8775 | Cult Movies, Dramas, Thrillers                 | 158 |
| 8776 | Kids' TV, Korean TV Shows, TV Comedies         | 2   |
| 8777 | Comedies, Horror Movies                        | 88  |
| 8778 | Children & Family Movies, Comedies             | 88  |
| 8779 | Dramas, International Movies, Music & Musicals | 111 |

```

duration_Str
0          min
1      Seasons
2      Season
3      Season
4      Seasons
...
8775      min
8776      Seasons
8777      min
8778      min
8779      min

```

[8780 rows x 12 columns]

*#j) split date\_added column into 3 separate cols*

```

dfj=df.copy()
dfj[['month', 'day', 'year']] =
dfj['date_added'].str.split(expand=True)
dfj['day'] = dfj['day'].str.replace(',', '', regex=False)
dfj.drop('date_added', axis=1,inplace=True)
dfj

```

|      | show_id | type    | title                 | director \      |
|------|---------|---------|-----------------------|-----------------|
| 0    | s1      | Movie   | Dick Johnson Is Dead  | Kirsten Johnson |
| 1    | s2      | TV Show | Blood & Water         | NaN             |
| 2    | s3      | TV Show | Ganglands             | Julien Leclercq |
| 3    | s4      | TV Show | Jailbirds New Orleans | NaN             |
| 4    | s5      | TV Show | Kota Factory          | NaN             |
| ...  | ...     | ...     | ...                   | ...             |
| 8775 | s8803   | Movie   | Zodiac                | David Fincher   |
| 8776 | s8804   | TV Show | Zombie Dumb           | NaN             |
| 8777 | s8805   | Movie   | Zombieland            | Ruben Fleischer |
| 8778 | s8806   | Movie   | Zoom                  | Peter Hewitt    |
| 8779 | s8807   | Movie   | Zubaan                | Mozez Singh     |

|   | cast | country       |
|---|------|---------------|
| \ |      |               |
| 0 | NaN  | United States |

|      |                                                   |               |
|------|---------------------------------------------------|---------------|
| 1    | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... | South Africa  |
| 2    | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... | NaN           |
| 3    |                                                   | NaN           |
| 4    | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... | India         |
| ...  | ...                                               | ...           |
| 8775 | Mark Ruffalo, Jake Gyllenhaal, Robert Downey J... | United States |
| 8776 |                                                   | NaN           |
| 8777 | Jesse Eisenberg, Woody Harrelson, Emma Stone, ... | United States |
| 8778 | Tim Allen, Courteney Cox, Chevy Chase, Kate Ma... | United States |
| 8779 | Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan... | India         |

|      | release_year | rating | duration \ |
|------|--------------|--------|------------|
| 0    | 2020         | PG-13  | 90 min     |
| 1    | 2021         | TV-MA  | 2 Seasons  |
| 2    | 2021         | TV-MA  | 1 Season   |
| 3    | 2021         | TV-MA  | 1 Season   |
| 4    | 2021         | TV-MA  | 2 Seasons  |
| ...  | ...          | ...    | ...        |
| 8775 | 2007         | R      | 158 min    |
| 8776 | 2018         | TV-Y7  | 2 Seasons  |
| 8777 | 2009         | R      | 88 min     |
| 8778 | 2006         | PG     | 88 min     |
| 8779 | 2015         | TV-14  | 111 min    |

|      | listed_in                                         | month     | day |
|------|---------------------------------------------------|-----------|-----|
| year |                                                   |           |     |
| 0    | Documentaries                                     | September | 25  |
| 2021 |                                                   |           |     |
| 1    | International TV Shows, TV Dramas, TV Mysteries   | September | 24  |
| 2021 |                                                   |           |     |
| 2    | Crime TV Shows, International TV Shows, TV Act... | September | 24  |
| 2021 |                                                   |           |     |
| 3    | Docuseries, Reality TV                            | September | 24  |
| 2021 |                                                   |           |     |
| 4    | International TV Shows, Romantic TV Shows, TV ... | September | 24  |
| 2021 |                                                   |           |     |
| ...  | ...                                               | ...       | ..  |
| ...  |                                                   |           |     |
| 8775 | Cult Movies, Dramas, Thrillers                    | November  | 20  |

|      |                                                |          |    |  |
|------|------------------------------------------------|----------|----|--|
| 2019 |                                                |          |    |  |
| 8776 | Kids' TV, Korean TV Shows, TV Comedies         | July     | 1  |  |
| 2019 |                                                |          |    |  |
| 8777 | Comedies, Horror Movies                        | November | 1  |  |
| 2019 |                                                |          |    |  |
| 8778 | Children & Family Movies, Comedies             | January  | 11 |  |
| 2020 |                                                |          |    |  |
| 8779 | Dramas, International Movies, Music & Musicals | March    | 2  |  |
| 2019 |                                                |          |    |  |

[8780 rows x 13 columns]

*#k) count of TV shows released per year*

```
release_count = df['release_year'].value_counts().sort_index()
release_count
```

```
release_year
1925      1
1942      2
1943      3
1944      3
1945      4
...
2017    1030
2018    1141
2019    1024
2020     952
2021     592
```

Name: count, Length: 74, dtype: int64

*#l)Rename column - title to movie\_title*

```
dfl=df.copy()
dfl.rename(columns={'title': 'movie_title'}, inplace=True)
dfl
```

|      | show_id | type    | movie_title           | director        | \ |
|------|---------|---------|-----------------------|-----------------|---|
| 0    | s1      | Movie   | Dick Johnson Is Dead  | Kirsten Johnson |   |
| 1    | s2      | TV Show | Blood & Water         | NaN             |   |
| 2    | s3      | TV Show | Ganglands             | Julien Leclercq |   |
| 3    | s4      | TV Show | Jailbirds New Orleans | NaN             |   |
| 4    | s5      | TV Show | Kota Factory          | NaN             |   |
| ...  | ...     | ...     | ...                   | ...             |   |
| 8775 | s8803   | Movie   | Zodiac                | David Fincher   |   |
| 8776 | s8804   | TV Show | Zombie Dumb           | NaN             |   |
| 8777 | s8805   | Movie   | Zombieland            | Ruben Fleischer |   |
| 8778 | s8806   | Movie   | Zoom                  | Peter Hewitt    |   |
| 8779 | s8807   | Movie   | Zubaan                | Mozez Singh     |   |

|   | cast | country |
|---|------|---------|
| \ |      |         |

|                          |                                                   |               |                   |
|--------------------------|---------------------------------------------------|---------------|-------------------|
| 0                        |                                                   | NaN           | United States     |
| 1                        | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... |               | South Africa      |
| 2                        | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... |               | NaN               |
| 3                        |                                                   | NaN           | NaN               |
| 4                        | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... |               | India             |
| ...                      | ...                                               |               | ...               |
| 8775                     | Mark Ruffalo, Jake Gyllenhaal, Robert Downey J... |               | United States     |
| 8776                     |                                                   | NaN           | NaN               |
| 8777                     | Jesse Eisenberg, Woody Harrelson, Emma Stone, ... |               | United States     |
| 8778                     | Tim Allen, Courteney Cox, Chevy Chase, Kate Ma... |               | United States     |
| 8779                     | Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan... |               | India             |
|                          |                                                   |               |                   |
|                          | date_added                                        | release_year  | rating duration \ |
| 0                        | September 25, 2021                                | 2020          | PG-13 90 min      |
| 1                        | September 24, 2021                                | 2021          | TV-MA 2 Seasons   |
| 2                        | September 24, 2021                                | 2021          | TV-MA 1 Season    |
| 3                        | September 24, 2021                                | 2021          | TV-MA 1 Season    |
| 4                        | September 24, 2021                                | 2021          | TV-MA 2 Seasons   |
| ...                      | ...                                               | ...           | ...               |
| 8775                     | November 20, 2019                                 | 2007          | R 158 min         |
| 8776                     | July 1, 2019                                      | 2018          | TV-Y7 2 Seasons   |
| 8777                     | November 1, 2019                                  | 2009          | R 88 min          |
| 8778                     | January 11, 2020                                  | 2006          | PG 88 min         |
| 8779                     | March 2, 2019                                     | 2015          | TV-14 111 min     |
|                          |                                                   |               |                   |
|                          |                                                   | listed_in     |                   |
| 0                        |                                                   | Documentaries |                   |
| 1                        | International TV Shows, TV Dramas, TV Mysteries   |               |                   |
| 2                        | Crime TV Shows, International TV Shows, TV Act... |               |                   |
| 3                        | Docuseries, Reality TV                            |               |                   |
| 4                        | International TV Shows, Romantic TV Shows, TV ... |               |                   |
| ...                      |                                                   | ...           |                   |
| 8775                     | Cult Movies, Dramas, Thrillers                    |               |                   |
| 8776                     | Kids' TV, Korean TV Shows, TV Comedies            |               |                   |
| 8777                     | Comedies, Horror Movies                           |               |                   |
| 8778                     | Children & Family Movies, Comedies                |               |                   |
| 8779                     | Dramas, International Movies, Music & Musicals    |               |                   |
|                          |                                                   |               |                   |
| [8780 rows x 11 columns] |                                                   |               |                   |

```
#m) split listed_in column to 3 separate cols - Genre1,Genre2, Genre3
dfm=df.copy()
dfm[['Genre1', 'Genre2', 'Genre3']] = dfm['listed_in'].str.split(',',
expand=True, n=2)
dfm.drop('listed_in',axis=1,inplace=True)
dfm
```

|      | show_id | type    | title                 | director        | \ |
|------|---------|---------|-----------------------|-----------------|---|
| 0    | s1      | Movie   | Dick Johnson Is Dead  | Kirsten Johnson |   |
| 1    | s2      | TV Show | Blood & Water         | NaN             |   |
| 2    | s3      | TV Show | Ganglands             | Julien Leclercq |   |
| 3    | s4      | TV Show | Jailbirds New Orleans | NaN             |   |
| 4    | s5      | TV Show | Kota Factory          | NaN             |   |
| ...  | ...     | ...     | ...                   | ...             |   |
| 8775 | s8803   | Movie   | Zodiac                | David Fincher   |   |
| 8776 | s8804   | TV Show | Zombie Dumb           | NaN             |   |
| 8777 | s8805   | Movie   | Zombieland            | Ruben Fleischer |   |
| 8778 | s8806   | Movie   | Zoom                  | Peter Hewitt    |   |
| 8779 | s8807   | Movie   | Zubaan                | Mozez Singh     |   |

|      | cast                                              | country       |
|------|---------------------------------------------------|---------------|
| \    |                                                   |               |
| 0    | NaN                                               | United States |
| 1    | Ama Qamata, Khosi Ngema, Gail Mablane, Thaban...  | South Africa  |
| 2    | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... | NaN           |
| 3    | NaN                                               | NaN           |
| 4    | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... | India         |
| ...  | ...                                               | ...           |
| 8775 | Mark Ruffalo, Jake Gyllenhaal, Robert Downey J... | United States |
| 8776 | NaN                                               | NaN           |
| 8777 | Jesse Eisenberg, Woody Harrelson, Emma Stone, ... | United States |
| 8778 | Tim Allen, Courteney Cox, Chevy Chase, Kate Ma... | United States |
| 8779 | Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan... | India         |

|   | date_added         | release_year | rating | duration  | \ |
|---|--------------------|--------------|--------|-----------|---|
| 0 | September 25, 2021 | 2020         | PG-13  | 90 min    |   |
| 1 | September 24, 2021 | 2021         | TV-MA  | 2 Seasons |   |
| 2 | September 24, 2021 | 2021         | TV-MA  | 1 Season  |   |
| 3 | September 24, 2021 | 2021         | TV-MA  | 1 Season  |   |
| 4 | September 24, 2021 | 2021         | TV-MA  | 2 Seasons |   |

|      |                   |      |       |           |
|------|-------------------|------|-------|-----------|
| ...  | ...               | ...  | ...   | ...       |
| 8775 | November 20, 2019 | 2007 | R     | 158 min   |
| 8776 | July 1, 2019      | 2018 | TV-Y7 | 2 Seasons |
| 8777 | November 1, 2019  | 2009 | R     | 88 min    |
| 8778 | January 11, 2020  | 2006 | PG    | 88 min    |
| 8779 | March 2, 2019     | 2015 | TV-14 | 111 min   |

|      |                          |                        |
|------|--------------------------|------------------------|
|      | Genre1                   | Genre2 \               |
| 0    | Documentaries            | None                   |
| 1    | International TV Shows   | TV Dramas              |
| 2    | Crime TV Shows           | International TV Shows |
| 3    | Docuseries               | Reality TV             |
| 4    | International TV Shows   | Romantic TV Shows      |
| ...  | ...                      | ...                    |
| 8775 | Cult Movies              | Dramas                 |
| 8776 | Kids' TV                 | Korean TV Shows        |
| 8777 | Comedies                 | Horror Movies          |
| 8778 | Children & Family Movies | Comedies               |
| 8779 | Dramas                   | International Movies   |

|      |                       |
|------|-----------------------|
|      | Genre3                |
| 0    | None                  |
| 1    | TV Mysteries          |
| 2    | TV Action & Adventure |
| 3    | None                  |
| 4    | TV Comedies           |
| ...  | ...                   |
| 8775 | Thrillers             |
| 8776 | TV Comedies           |
| 8777 | None                  |
| 8778 | None                  |
| 8779 | Music & Musicals      |

[8780 rows x 13 columns]

*#Q2. Import the student csv data to Dataframe*

```
import numpy as np
import pandas as pd
pd.set_option('display.max_rows', None)
dfs =
pd.read_csv('C:/Users/Admin/OneDrive/Desktop/IoT_Academy/Assignments/
Tests/student.csv')
dfs
```

|   | Student_id | Age | Gender | Grade     | Employed |
|---|------------|-----|--------|-----------|----------|
| 0 | 1          | 19  | Male   | 1st Class | yes      |
| 1 | 2          | 20  | Female | 2nd Class | no       |
| 2 | 3          | 18  | Male   | 1st Class | no       |
| 3 | 4          | 21  | Female | 2nd Class | no       |
| 4 | 5          | 19  | Male   | 1st Class | no       |



|    |    |    |        |           |     |
|----|----|----|--------|-----------|-----|
| 5  | 6  | 20 | Male   | 2nd Class | yes |
| 6  | 7  | 19 | Female | 3rd Class | yes |
| 7  | 8  | 21 | Male   | 3rd Class | yes |
| 8  | 9  | 22 | Female | 3rd Class | yes |
| 9  | 10 | 21 | Male   | 1st Class | no  |
| 10 | 11 | 20 | Female | 3rd Class | yes |
| 11 | 12 | 20 | Male   | 1st Class | no  |
| 12 | 13 | 19 | Male   | 1st Class | yes |
| 13 | 14 | 20 | Female | 3rd Class | no  |
| 14 | 15 | 19 | Male   | 1st Class | yes |
| 15 | 16 | 19 | Female | 2nd Class | no  |
| 16 | 17 | 20 | Male   | 2nd Class | yes |
| 17 | 18 | 18 | Female | 3rd Class | no  |
| 18 | 19 | 21 | Male   | 2nd Class | yes |
| 19 | 20 | 19 | Male   | 2nd Class | no  |
| 20 | 21 | 20 | Female | 3rd Class | yes |
| 21 | 22 | 19 | Male   | 2nd Class | no  |
| 22 | 23 | 21 | Female | 2nd Class | yes |
| 23 | 24 | 22 | Male   | 3rd Class | no  |
| 24 | 25 | 21 | Female | 1st Class | no  |
| 25 | 26 | 20 | Male   | 2nd Class | no  |
| 26 | 27 | 20 | Male   | 3rd Class | no  |
| 27 | 28 | 19 | Female | 1st Class | no  |
| 28 | 29 | 20 | Male   | 3rd Class | no  |
| 29 | 30 | 19 | Male   | 1st Class | yes |
| 30 | 31 | 20 | Female | 2nd Class | no  |
| 31 | 32 | 18 | Male   | 1st Class | no  |
| 32 | 33 | 21 | Female | 2nd Class | no  |
| 33 | 34 | 19 | Male   | 1st Class | no  |
| 34 | 35 | 20 | Male   | 2nd Class | yes |
| 35 | 36 | 19 | Female | 3rd Class | yes |
| 36 | 37 | 21 | Male   | 3rd Class | yes |
| 37 | 38 | 22 | Female | 3rd Class | yes |
| 38 | 39 | 21 | Male   | 1st Class | no  |
| 39 | 40 | 20 | Female | 3rd Class | yes |
| 40 | 41 | 20 | Male   | 1st Class | no  |
| 41 | 42 | 19 | Male   | 1st Class | yes |
| 42 | 43 | 20 | Female | 3rd Class | no  |
| 43 | 44 | 19 | Male   | 1st Class | yes |
| 44 | 45 | 19 | Female | 2nd Class | no  |
| 45 | 46 | 20 | Male   | 2nd Class | yes |
| 46 | 47 | 18 | Female | 3rd Class | no  |
| 47 | 48 | 21 | Male   | 2nd Class | yes |
| 48 | 49 | 19 | Male   | 2nd Class | no  |
| 49 | 50 | 20 | Female | 3rd Class | yes |
| 50 | 51 | 19 | Male   | 2nd Class | no  |
| 51 | 52 | 21 | Female | 2nd Class | yes |
| 52 | 53 | 22 | Male   | 3rd Class | no  |
| 53 | 54 | 21 | Female | 1st Class | no  |

|     |     |    |        |           |     |
|-----|-----|----|--------|-----------|-----|
| 54  | 55  | 20 | Male   | 2nd Class | no  |
| 55  | 56  | 20 | Male   | 3rd Class | no  |
| 56  | 57  | 19 | Female | 1st Class | no  |
| 57  | 58  | 20 | Male   | 3rd Class | no  |
| 58  | 59  | 19 | Male   | 1st Class | yes |
| 59  | 60  | 20 | Female | 2nd Class | no  |
| 60  | 61  | 18 | Male   | 1st Class | no  |
| 61  | 62  | 21 | Female | 2nd Class | no  |
| 62  | 63  | 19 | Male   | 1st Class | no  |
| 63  | 64  | 20 | Male   | 2nd Class | yes |
| 64  | 65  | 19 | Female | 3rd Class | yes |
| 65  | 66  | 21 | Male   | 3rd Class | yes |
| 66  | 67  | 22 | Female | 3rd Class | yes |
| 67  | 68  | 21 | Male   | 1st Class | no  |
| 68  | 69  | 20 | Female | 3rd Class | yes |
| 69  | 70  | 20 | Male   | 1st Class | no  |
| 70  | 71  | 19 | Male   | 1st Class | yes |
| 71  | 72  | 20 | Female | 3rd Class | no  |
| 72  | 73  | 19 | Male   | 1st Class | yes |
| 73  | 74  | 19 | Female | 2nd Class | no  |
| 74  | 75  | 20 | Male   | 2nd Class | yes |
| 75  | 76  | 18 | Female | 3rd Class | no  |
| 76  | 77  | 21 | Male   | 2nd Class | yes |
| 77  | 78  | 19 | Male   | 2nd Class | no  |
| 78  | 79  | 20 | Female | 3rd Class | yes |
| 79  | 80  | 19 | Male   | 2nd Class | no  |
| 80  | 81  | 21 | Female | 2nd Class | yes |
| 81  | 82  | 22 | Male   | 3rd Class | no  |
| 82  | 83  | 21 | Female | 1st Class | no  |
| 83  | 84  | 20 | Male   | 2nd Class | no  |
| 84  | 85  | 20 | Male   | 3rd Class | no  |
| 85  | 86  | 19 | Female | 1st Class | no  |
| 86  | 87  | 20 | Male   | 3rd Class | no  |
| 87  | 88  | 19 | Male   | 1st Class | yes |
| 88  | 89  | 20 | Female | 2nd Class | no  |
| 89  | 90  | 18 | Male   | 1st Class | no  |
| 90  | 91  | 21 | Female | 2nd Class | no  |
| 91  | 92  | 19 | Male   | 1st Class | no  |
| 92  | 93  | 20 | Male   | 2nd Class | yes |
| 93  | 94  | 19 | Female | 3rd Class | yes |
| 94  | 95  | 21 | Male   | 3rd Class | yes |
| 95  | 96  | 22 | Female | 3rd Class | yes |
| 96  | 97  | 21 | Male   | 1st Class | no  |
| 97  | 98  | 20 | Female | 3rd Class | yes |
| 98  | 99  | 20 | Male   | 1st Class | no  |
| 99  | 100 | 19 | Male   | 1st Class | yes |
| 100 | 101 | 20 | Female | 3rd Class | no  |
| 101 | 102 | 19 | Male   | 1st Class | yes |
| 102 | 103 | 19 | Female | 2nd Class | no  |

|     |     |    |        |           |     |
|-----|-----|----|--------|-----------|-----|
| 103 | 104 | 20 | Male   | 2nd Class | yes |
| 104 | 105 | 18 | Female | 3rd Class | no  |
| 105 | 106 | 21 | Male   | 2nd Class | yes |
| 106 | 107 | 19 | Male   | 2nd Class | no  |
| 107 | 108 | 20 | Female | 3rd Class | yes |
| 108 | 109 | 19 | Male   | 2nd Class | no  |
| 109 | 110 | 21 | Female | 2nd Class | yes |
| 110 | 111 | 22 | Male   | 3rd Class | no  |
| 111 | 112 | 21 | Female | 1st Class | no  |
| 112 | 113 | 20 | Male   | 2nd Class | no  |
| 113 | 114 | 20 | Male   | 3rd Class | no  |
| 114 | 115 | 19 | Female | 1st Class | no  |
| 115 | 116 | 20 | Male   | 3rd Class | no  |
| 116 | 117 | 19 | Male   | 1st Class | yes |
| 117 | 118 | 20 | Female | 2nd Class | no  |
| 118 | 119 | 18 | Male   | 1st Class | no  |
| 119 | 120 | 21 | Female | 2nd Class | no  |
| 120 | 121 | 19 | Male   | 1st Class | no  |
| 121 | 122 | 20 | Male   | 2nd Class | yes |
| 122 | 123 | 19 | Female | 3rd Class | yes |
| 123 | 124 | 21 | Male   | 3rd Class | yes |
| 124 | 125 | 22 | Female | 3rd Class | yes |
| 125 | 126 | 21 | Male   | 1st Class | no  |
| 126 | 127 | 20 | Female | 3rd Class | yes |
| 127 | 128 | 20 | Male   | 1st Class | no  |
| 128 | 129 | 19 | Male   | 1st Class | yes |
| 129 | 130 | 20 | Female | 3rd Class | no  |
| 130 | 131 | 19 | Male   | 1st Class | yes |
| 131 | 132 | 19 | Female | 2nd Class | no  |
| 132 | 133 | 20 | Male   | 2nd Class | yes |
| 133 | 134 | 18 | Female | 3rd Class | no  |
| 134 | 135 | 21 | Male   | 2nd Class | yes |
| 135 | 136 | 19 | Male   | 2nd Class | no  |
| 136 | 137 | 20 | Female | 3rd Class | yes |
| 137 | 138 | 19 | Male   | 2nd Class | no  |
| 138 | 139 | 21 | Female | 2nd Class | yes |
| 139 | 140 | 22 | Male   | 3rd Class | no  |
| 140 | 141 | 21 | Female | 1st Class | no  |
| 141 | 142 | 20 | Male   | 2nd Class | no  |
| 142 | 143 | 20 | Male   | 3rd Class | no  |
| 143 | 144 | 19 | Female | 1st Class | no  |
| 144 | 145 | 20 | Male   | 3rd Class | no  |
| 145 | 146 | 19 | Male   | 1st Class | yes |
| 146 | 147 | 20 | Female | 2nd Class | no  |
| 147 | 148 | 18 | Male   | 1st Class | no  |
| 148 | 149 | 21 | Female | 2nd Class | no  |
| 149 | 150 | 19 | Male   | 1st Class | no  |
| 150 | 151 | 20 | Male   | 2nd Class | yes |
| 151 | 152 | 19 | Female | 3rd Class | yes |

|     |     |    |        |           |     |
|-----|-----|----|--------|-----------|-----|
| 152 | 153 | 21 | Male   | 3rd Class | yes |
| 153 | 154 | 22 | Female | 3rd Class | yes |
| 154 | 155 | 21 | Male   | 1st Class | no  |
| 155 | 156 | 20 | Female | 3rd Class | yes |
| 156 | 157 | 20 | Male   | 1st Class | no  |
| 157 | 158 | 19 | Male   | 1st Class | yes |
| 158 | 159 | 20 | Female | 3rd Class | no  |
| 159 | 160 | 19 | Male   | 1st Class | yes |
| 160 | 161 | 19 | Female | 2nd Class | no  |
| 161 | 162 | 20 | Male   | 2nd Class | yes |
| 162 | 163 | 18 | Female | 3rd Class | no  |
| 163 | 164 | 21 | Male   | 2nd Class | yes |
| 164 | 165 | 19 | Male   | 2nd Class | no  |
| 165 | 166 | 20 | Female | 3rd Class | yes |
| 166 | 167 | 19 | Male   | 2nd Class | no  |
| 167 | 168 | 21 | Female | 2nd Class | yes |
| 168 | 169 | 22 | Male   | 3rd Class | no  |
| 169 | 170 | 21 | Female | 1st Class | no  |
| 170 | 171 | 20 | Male   | 2nd Class | no  |
| 171 | 172 | 20 | Male   | 3rd Class | no  |
| 172 | 173 | 19 | Female | 1st Class | no  |
| 173 | 174 | 20 | Male   | 3rd Class | no  |
| 174 | 175 | 19 | Male   | 1st Class | yes |
| 175 | 176 | 20 | Female | 2nd Class | no  |
| 176 | 177 | 18 | Male   | 1st Class | no  |
| 177 | 178 | 21 | Female | 2nd Class | no  |
| 178 | 179 | 19 | Male   | 1st Class | no  |
| 179 | 180 | 20 | Male   | 2nd Class | yes |
| 180 | 181 | 19 | Female | 3rd Class | yes |
| 181 | 182 | 21 | Male   | 3rd Class | yes |
| 182 | 183 | 22 | Female | 3rd Class | yes |
| 183 | 184 | 21 | Male   | 1st Class | no  |
| 184 | 185 | 20 | Female | 3rd Class | yes |
| 185 | 186 | 20 | Male   | 1st Class | no  |
| 186 | 187 | 19 | Male   | 1st Class | yes |
| 187 | 188 | 20 | Female | 3rd Class | no  |
| 188 | 189 | 19 | Male   | 1st Class | yes |
| 189 | 190 | 19 | Female | 2nd Class | no  |
| 190 | 191 | 20 | Male   | 2nd Class | yes |
| 191 | 192 | 18 | Female | 3rd Class | no  |
| 192 | 193 | 21 | Male   | 2nd Class | yes |
| 193 | 194 | 19 | Male   | 2nd Class | no  |
| 194 | 195 | 20 | Female | 3rd Class | yes |
| 195 | 196 | 19 | Male   | 2nd Class | no  |
| 196 | 197 | 21 | Female | 2nd Class | yes |
| 197 | 198 | 22 | Male   | 3rd Class | no  |
| 198 | 199 | 21 | Female | 1st Class | no  |
| 199 | 200 | 20 | Male   | 2nd Class | no  |
| 200 | 201 | 20 | Male   | 3rd Class | no  |

|     |     |    |        |           |     |
|-----|-----|----|--------|-----------|-----|
| 201 | 202 | 19 | Female | 1st Class | no  |
| 202 | 203 | 20 | Male   | 3rd Class | no  |
| 203 | 204 | 19 | Male   | 1st Class | yes |
| 204 | 205 | 20 | Female | 2nd Class | no  |
| 205 | 206 | 18 | Male   | 1st Class | no  |
| 206 | 207 | 21 | Female | 2nd Class | no  |
| 207 | 208 | 19 | Male   | 1st Class | no  |
| 208 | 209 | 20 | Male   | 2nd Class | yes |
| 209 | 210 | 19 | Female | 3rd Class | yes |
| 210 | 211 | 21 | Male   | 3rd Class | yes |
| 211 | 212 | 22 | Female | 3rd Class | yes |
| 212 | 213 | 21 | Male   | 1st Class | no  |
| 213 | 214 | 20 | Female | 3rd Class | yes |
| 214 | 215 | 20 | Male   | 1st Class | no  |
| 215 | 216 | 19 | Male   | 1st Class | yes |
| 216 | 217 | 20 | Female | 3rd Class | no  |
| 217 | 218 | 19 | Male   | 1st Class | yes |
| 218 | 219 | 19 | Female | 2nd Class | no  |
| 219 | 220 | 20 | Male   | 2nd Class | yes |
| 220 | 221 | 18 | Female | 3rd Class | no  |
| 221 | 222 | 21 | Male   | 2nd Class | yes |
| 222 | 223 | 19 | Male   | 2nd Class | no  |
| 223 | 224 | 20 | Female | 3rd Class | yes |
| 224 | 225 | 19 | Male   | 2nd Class | no  |
| 225 | 226 | 21 | Female | 2nd Class | yes |
| 226 | 227 | 22 | Male   | 3rd Class | no  |
| 227 | 228 | 21 | Female | 1st Class | no  |
| 228 | 229 | 20 | Male   | 2nd Class | no  |
| 229 | 230 | 20 | Male   | 3rd Class | yes |
| 230 | 231 | 19 | Female | 1st Class | yes |
| 231 | 232 | 20 | Male   | 3rd Class | yes |

*#Q2. Import the mark csv data to Dataframe*

```
dfm=
```

```
pd.read_csv('C:/Users/Admin/OneDrive/Desktop/IoT_Academy/Assignments/Tests/mark.csv')
```

```
dfm
```

|    | Student_id | Mark | City    |
|----|------------|------|---------|
| 0  | 1          | 95   | Chennai |
| 1  | 2          | 70   | Delhi   |
| 2  | 3          | 98   | Mumbai  |
| 3  | 4          | 75   | Pune    |
| 4  | 5          | 89   | Kochi   |
| 5  | 6          | 69   | Gwalior |
| 6  | 7          | 52   | Bhopal  |
| 7  | 8          | 54   | Chennai |
| 8  | 11         | 52   | Pune    |
| 9  | 12         | 90   | Kochi   |
| 10 | 13         | 92   | Gwalior |

|    |    |    |         |
|----|----|----|---------|
| 11 | 14 | 55 | Bhopal  |
| 12 | 15 | 97 | Pune    |
| 13 | 16 | 77 | Kochi   |
| 14 | 17 | 68 | Gwalior |
| 15 | 18 | 51 | Bhopal  |
| 16 | 19 | 72 | Chennai |
| 17 | 20 | 79 | Delhi   |
| 18 | 21 | 49 | Mumbai  |
| 19 | 22 | 78 | Pune    |
| 20 | 23 | 76 | Chennai |
| 21 | 24 | 43 | Delhi   |
| 22 | 25 | 89 | Mumbai  |
| 23 | 26 | 75 | Pune    |
| 24 | 27 | 45 | Kochi   |
| 25 | 28 | 98 | Gwalior |
| 26 | 29 | 54 | Bhopal  |
| 27 | 30 | 81 | Chennai |
| 28 | 31 | 74 | Delhi   |
| 29 | 32 | 82 | Mumbai  |
| 30 | 33 | 78 | Pune    |
| 31 | 34 | 83 | Kochi   |
| 32 | 35 | 79 | Gwalior |
| 33 | 36 | 55 | Bhopal  |
| 34 | 37 | 53 | Pune    |
| 35 | 38 | 55 | Kochi   |
| 36 | 39 | 84 | Gwalior |
| 37 | 40 | 44 | Bhopal  |
| 38 | 41 | 85 | Chennai |
| 39 | 42 | 86 | Delhi   |
| 40 | 43 | 45 | Mumbai  |
| 41 | 44 | 95 | Pune    |
| 42 | 45 | 73 | Chennai |
| 43 | 46 | 79 | Delhi   |
| 44 | 47 | 46 | Mumbai  |
| 45 | 48 | 67 | Pune    |
| 46 | 50 | 47 | Gwalior |
| 47 | 51 | 65 | Bhopal  |
| 48 | 52 | 67 | Chennai |
| 49 | 53 | 49 | Delhi   |
| 50 | 54 | 82 | Mumbai  |
| 51 | 55 | 69 | Pune    |
| 52 | 56 | 48 | Kochi   |
| 53 | 57 | 85 | Gwalior |
| 54 | 58 | 42 | Bhopal  |
| 55 | 59 | 89 | Pune    |
| 56 | 60 | 66 | Kochi   |
| 57 | 61 | 98 | Gwalior |
| 58 | 62 | 77 | Bhopal  |
| 59 | 63 | 98 | Chennai |

|     |     |    |         |
|-----|-----|----|---------|
| 60  | 64  | 78 | Delhi   |
| 61  | 65  | 43 | Mumbai  |
| 62  | 66  | 59 | Pune    |
| 63  | 67  | 58 | Chennai |
| 64  | 68  | 87 | Delhi   |
| 65  | 69  | 55 | Mumbai  |
| 66  | 70  | 86 | Pune    |
| 67  | 71  | 85 | Kochi   |
| 68  | 72  | 54 | Gwalior |
| 69  | 73  | 89 | Bhopal  |
| 70  | 74  | 77 | Chennai |
| 71  | 75  | 79 | Delhi   |
| 72  | 76  | 53 | Mumbai  |
| 73  | 77  | 77 | Pune    |
| 74  | 78  | 76 | Kochi   |
| 75  | 79  | 56 | Gwalior |
| 76  | 80  | 75 | Bhopal  |
| 77  | 81  | 74 | Pune    |
| 78  | 82  | 59 | Kochi   |
| 79  | 83  | 98 | Gwalior |
| 80  | 84  | 73 | Bhopal  |
| 81  | 85  | 58 | Chennai |
| 82  | 86  | 97 | Delhi   |
| 83  | 87  | 57 | Mumbai  |
| 84  | 88  | 96 | Pune    |
| 85  | 89  | 75 | Chennai |
| 86  | 90  | 94 | Delhi   |
| 87  | 91  | 74 | Mumbai  |
| 88  | 92  | 93 | Pune    |
| 89  | 93  | 74 | Kochi   |
| 90  | 94  | 53 | Gwalior |
| 91  | 95  | 50 | Bhopal  |
| 92  | 96  | 40 | Chennai |
| 93  | 97  | 87 | Delhi   |
| 94  | 98  | 44 | Mumbai  |
| 95  | 99  | 86 | Pune    |
| 96  | 100 | 84 | Kochi   |
| 97  | 101 | 43 | Gwalior |
| 98  | 102 | 98 | Chennai |
| 99  | 103 | 75 | Delhi   |
| 100 | 104 | 78 | Mumbai  |
| 101 | 105 | 42 | Pune    |
| 102 | 106 | 77 | Kochi   |
| 103 | 107 | 78 | Gwalior |
| 104 | 108 | 41 | Bhopal  |
| 105 | 109 | 79 | Chennai |
| 106 | 110 | 76 | Delhi   |
| 107 | 111 | 45 | Mumbai  |
| 108 | 112 | 97 | Pune    |

|     |     |     |         |
|-----|-----|-----|---------|
| 109 | 113 | 75  | Kochi   |
| 110 | 114 | 46  | Gwalior |
| 111 | 115 | 96  | Bhopal  |
| 112 | 116 | 47  | Pune    |
| 113 | 117 | 90  | Kochi   |
| 114 | 118 | 79  | Gwalior |
| 115 | 119 | 91  | Bhopal  |
| 116 | 120 | 77  | Chennai |
| 117 | 121 | 87  | Delhi   |
| 118 | 122 | 75  | Mumbai  |
| 119 | 123 | 48  | Pune    |
| 120 | 124 | 49  | Chennai |
| 121 | 125 | 55  | Delhi   |
| 122 | 126 | 88  | Mumbai  |
| 123 | 127 | 56  | Pune    |
| 124 | 128 | 86  | Kochi   |
| 125 | 129 | 89  | Gwalior |
| 126 | 130 | 57  | Bhopal  |
| 127 | 131 | 90  | Chennai |
| 128 | 132 | 75  | Delhi   |
| 129 | 133 | 75  | Mumbai  |
| 130 | 134 | 58  | Pune    |
| 131 | 135 | 75  | Kochi   |
| 132 | 136 | 74  | Gwalior |
| 133 | 137 | 59  | Chennai |
| 134 | 138 | 73  | Delhi   |
| 135 | 139 | 72  | Mumbai  |
| 136 | 140 | 50  | Pune    |
| 137 | 141 | 93  | Kochi   |
| 138 | 142 | 71  | Gwalior |
| 139 | 143 | 40  | Bhopal  |
| 140 | 144 | 97  | Chennai |
| 141 | 145 | 43  | Delhi   |
| 142 | 146 | 98  | Mumbai  |
| 143 | 147 | 70  | Pune    |
| 144 | 148 | 99  | Kochi   |
| 145 | 149 | 77  | Gwalior |
| 146 | 150 | 100 | Bhopal  |
| 147 | 151 | 78  | Pune    |
| 148 | 152 | 45  | Kochi   |
| 149 | 153 | 46  | Gwalior |
| 150 | 154 | 47  | Bhopal  |
| 151 | 155 | 89  | Chennai |
| 152 | 156 | 49  | Delhi   |
| 153 | 157 | 86  | Mumbai  |
| 154 | 158 | 92  | Pune    |
| 155 | 159 | 50  | Chennai |
| 156 | 160 | 92  | Delhi   |
| 157 | 161 | 79  | Mumbai  |



|     |     |    |         |
|-----|-----|----|---------|
| 158 | 162 | 79 | Pune    |
| 159 | 163 | 59 | Kochi   |
| 160 | 164 | 75 | Gwalior |
| 161 | 165 | 75 | Bhopal  |
| 162 | 166 | 53 | Chennai |
| 163 | 167 | 74 | Delhi   |
| 164 | 168 | 74 | Mumbai  |
| 165 | 169 | 53 | Pune    |
| 166 | 170 | 90 | Kochi   |
| 167 | 171 | 73 | Gwalior |
| 168 | 172 | 52 | Chennai |
| 169 | 173 | 98 | Delhi   |
| 170 | 174 | 56 | Mumbai  |
| 171 | 175 | 84 | Pune    |
| 172 | 176 | 77 | Kochi   |
| 173 | 177 | 97 | Gwalior |
| 174 | 178 | 76 | Bhopal  |
| 175 | 179 | 94 | Chennai |
| 176 | 180 | 76 | Delhi   |
| 177 | 181 | 56 | Mumbai  |
| 178 | 182 | 58 | Pune    |
| 179 | 183 | 59 | Kochi   |
| 180 | 184 | 87 | Gwalior |
| 181 | 185 | 50 | Bhopal  |
| 182 | 186 | 85 | Pune    |
| 183 | 187 | 83 | Kochi   |
| 184 | 188 | 40 | Gwalior |
| 185 | 189 | 89 | Bhopal  |
| 186 | 190 | 75 | Chennai |
| 187 | 191 | 77 | Delhi   |
| 188 | 192 | 44 | Mumbai  |
| 189 | 193 | 75 | Pune    |
| 190 | 194 | 74 | Chennai |
| 191 | 195 | 43 | Delhi   |
| 192 | 196 | 74 | Mumbai  |
| 193 | 197 | 74 | Pune    |
| 194 | 198 | 53 | Kochi   |
| 195 | 199 | 87 | Gwalior |
| 196 | 200 | 75 | Bhopal  |
| 197 | 201 | 42 | Chennai |
| 198 | 202 | 80 | Delhi   |
| 199 | 203 | 46 | Mumbai  |
| 200 | 204 | 91 | Pune    |
| 201 | 205 | 78 | Kochi   |
| 202 | 206 | 80 | Gwalior |
| 203 | 207 | 79 | Chennai |
| 204 | 208 | 91 | Delhi   |
| 205 | 209 | 70 | Mumbai  |
| 206 | 210 | 59 | Pune    |

|     |     |    |         |
|-----|-----|----|---------|
| 207 | 211 | 58 | Kochi   |
| 208 | 212 | 56 | Gwalior |
| 209 | 213 | 92 | Bhopal  |
| 210 | 214 | 54 | Chennai |
| 211 | 215 | 93 | Delhi   |
| 212 | 216 | 95 | Mumbai  |
| 213 | 217 | 53 | Pune    |
| 214 | 218 | 96 | Kochi   |
| 215 | 219 | 75 | Gwalior |
| 216 | 220 | 75 | Bhopal  |
| 217 | 221 | 52 | Pune    |
| 218 | 222 | 74 | Kochi   |
| 219 | 223 | 79 | Gwalior |
| 220 | 224 | 50 | Bhopal  |
| 221 | 225 | 72 | Chennai |
| 222 | 226 | 71 | Delhi   |
| 223 | 227 | 51 | Mumbai  |
| 224 | 228 | 99 | Pune    |
| 225 | 229 | 70 | Chennai |
| 226 | 230 | 55 | Delhi   |
| 227 | 231 | 97 | Mumbai  |
| 228 | 232 | 59 | Pune    |

##Q2.a) combining 2 dataframes

```
combined_df = pd.merge(dfs, dfm, on="Student_id")
combined_df
```

|    | Student_id | Age | Gender | Grade     | Employed | Mark | City    |
|----|------------|-----|--------|-----------|----------|------|---------|
| 0  | 1          | 19  | Male   | 1st Class | yes      | 95   | Chennai |
| 1  | 2          | 20  | Female | 2nd Class | no       | 70   | Delhi   |
| 2  | 3          | 18  | Male   | 1st Class | no       | 98   | Mumbai  |
| 3  | 4          | 21  | Female | 2nd Class | no       | 75   | Pune    |
| 4  | 5          | 19  | Male   | 1st Class | no       | 89   | Kochi   |
| 5  | 6          | 20  | Male   | 2nd Class | yes      | 69   | Gwalior |
| 6  | 7          | 19  | Female | 3rd Class | yes      | 52   | Bhopal  |
| 7  | 8          | 21  | Male   | 3rd Class | yes      | 54   | Chennai |
| 8  | 11         | 20  | Female | 3rd Class | yes      | 52   | Pune    |
| 9  | 12         | 20  | Male   | 1st Class | no       | 90   | Kochi   |
| 10 | 13         | 19  | Male   | 1st Class | yes      | 92   | Gwalior |
| 11 | 14         | 20  | Female | 3rd Class | no       | 55   | Bhopal  |
| 12 | 15         | 19  | Male   | 1st Class | yes      | 97   | Pune    |
| 13 | 16         | 19  | Female | 2nd Class | no       | 77   | Kochi   |
| 14 | 17         | 20  | Male   | 2nd Class | yes      | 68   | Gwalior |
| 15 | 18         | 18  | Female | 3rd Class | no       | 51   | Bhopal  |
| 16 | 19         | 21  | Male   | 2nd Class | yes      | 72   | Chennai |
| 17 | 20         | 19  | Male   | 2nd Class | no       | 79   | Delhi   |
| 18 | 21         | 20  | Female | 3rd Class | yes      | 49   | Mumbai  |
| 19 | 22         | 19  | Male   | 2nd Class | no       | 78   | Pune    |
| 20 | 23         | 21  | Female | 2nd Class | yes      | 76   | Chennai |
| 21 | 24         | 22  | Male   | 3rd Class | no       | 43   | Delhi   |

|    |    |    |        |           |     |    |         |
|----|----|----|--------|-----------|-----|----|---------|
| 22 | 25 | 21 | Female | 1st Class | no  | 89 | Mumbai  |
| 23 | 26 | 20 | Male   | 2nd Class | no  | 75 | Pune    |
| 24 | 27 | 20 | Male   | 3rd Class | no  | 45 | Kochi   |
| 25 | 28 | 19 | Female | 1st Class | no  | 98 | Gwalior |
| 26 | 29 | 20 | Male   | 3rd Class | no  | 54 | Bhopal  |
| 27 | 30 | 19 | Male   | 1st Class | yes | 81 | Chennai |
| 28 | 31 | 20 | Female | 2nd Class | no  | 74 | Delhi   |
| 29 | 32 | 18 | Male   | 1st Class | no  | 82 | Mumbai  |
| 30 | 33 | 21 | Female | 2nd Class | no  | 78 | Pune    |
| 31 | 34 | 19 | Male   | 1st Class | no  | 83 | Kochi   |
| 32 | 35 | 20 | Male   | 2nd Class | yes | 79 | Gwalior |
| 33 | 36 | 19 | Female | 3rd Class | yes | 55 | Bhopal  |
| 34 | 37 | 21 | Male   | 3rd Class | yes | 53 | Pune    |
| 35 | 38 | 22 | Female | 3rd Class | yes | 55 | Kochi   |
| 36 | 39 | 21 | Male   | 1st Class | no  | 84 | Gwalior |
| 37 | 40 | 20 | Female | 3rd Class | yes | 44 | Bhopal  |
| 38 | 41 | 20 | Male   | 1st Class | no  | 85 | Chennai |
| 39 | 42 | 19 | Male   | 1st Class | yes | 86 | Delhi   |
| 40 | 43 | 20 | Female | 3rd Class | no  | 45 | Mumbai  |
| 41 | 44 | 19 | Male   | 1st Class | yes | 95 | Pune    |
| 42 | 45 | 19 | Female | 2nd Class | no  | 73 | Chennai |
| 43 | 46 | 20 | Male   | 2nd Class | yes | 79 | Delhi   |
| 44 | 47 | 18 | Female | 3rd Class | no  | 46 | Mumbai  |
| 45 | 48 | 21 | Male   | 2nd Class | yes | 67 | Pune    |
| 46 | 50 | 20 | Female | 3rd Class | yes | 47 | Gwalior |
| 47 | 51 | 19 | Male   | 2nd Class | no  | 65 | Bhopal  |
| 48 | 52 | 21 | Female | 2nd Class | yes | 67 | Chennai |
| 49 | 53 | 22 | Male   | 3rd Class | no  | 49 | Delhi   |
| 50 | 54 | 21 | Female | 1st Class | no  | 82 | Mumbai  |
| 51 | 55 | 20 | Male   | 2nd Class | no  | 69 | Pune    |
| 52 | 56 | 20 | Male   | 3rd Class | no  | 48 | Kochi   |
| 53 | 57 | 19 | Female | 1st Class | no  | 85 | Gwalior |
| 54 | 58 | 20 | Male   | 3rd Class | no  | 42 | Bhopal  |
| 55 | 59 | 19 | Male   | 1st Class | yes | 89 | Pune    |
| 56 | 60 | 20 | Female | 2nd Class | no  | 66 | Kochi   |
| 57 | 61 | 18 | Male   | 1st Class | no  | 98 | Gwalior |
| 58 | 62 | 21 | Female | 2nd Class | no  | 77 | Bhopal  |
| 59 | 63 | 19 | Male   | 1st Class | no  | 98 | Chennai |
| 60 | 64 | 20 | Male   | 2nd Class | yes | 78 | Delhi   |
| 61 | 65 | 19 | Female | 3rd Class | yes | 43 | Mumbai  |
| 62 | 66 | 21 | Male   | 3rd Class | yes | 59 | Pune    |
| 63 | 67 | 22 | Female | 3rd Class | yes | 58 | Chennai |
| 64 | 68 | 21 | Male   | 1st Class | no  | 87 | Delhi   |
| 65 | 69 | 20 | Female | 3rd Class | yes | 55 | Mumbai  |
| 66 | 70 | 20 | Male   | 1st Class | no  | 86 | Pune    |
| 67 | 71 | 19 | Male   | 1st Class | yes | 85 | Kochi   |
| 68 | 72 | 20 | Female | 3rd Class | no  | 54 | Gwalior |
| 69 | 73 | 19 | Male   | 1st Class | yes | 89 | Bhopal  |
| 70 | 74 | 19 | Female | 2nd Class | no  | 77 | Chennai |

|     |     |    |        |           |     |    |         |
|-----|-----|----|--------|-----------|-----|----|---------|
| 71  | 75  | 20 | Male   | 2nd Class | yes | 79 | Delhi   |
| 72  | 76  | 18 | Female | 3rd Class | no  | 53 | Mumbai  |
| 73  | 77  | 21 | Male   | 2nd Class | yes | 77 | Pune    |
| 74  | 78  | 19 | Male   | 2nd Class | no  | 76 | Kochi   |
| 75  | 79  | 20 | Female | 3rd Class | yes | 56 | Gwalior |
| 76  | 80  | 19 | Male   | 2nd Class | no  | 75 | Bhopal  |
| 77  | 81  | 21 | Female | 2nd Class | yes | 74 | Pune    |
| 78  | 82  | 22 | Male   | 3rd Class | no  | 59 | Kochi   |
| 79  | 83  | 21 | Female | 1st Class | no  | 98 | Gwalior |
| 80  | 84  | 20 | Male   | 2nd Class | no  | 73 | Bhopal  |
| 81  | 85  | 20 | Male   | 3rd Class | no  | 58 | Chennai |
| 82  | 86  | 19 | Female | 1st Class | no  | 97 | Delhi   |
| 83  | 87  | 20 | Male   | 3rd Class | no  | 57 | Mumbai  |
| 84  | 88  | 19 | Male   | 1st Class | yes | 96 | Pune    |
| 85  | 89  | 20 | Female | 2nd Class | no  | 75 | Chennai |
| 86  | 90  | 18 | Male   | 1st Class | no  | 94 | Delhi   |
| 87  | 91  | 21 | Female | 2nd Class | no  | 74 | Mumbai  |
| 88  | 92  | 19 | Male   | 1st Class | no  | 93 | Pune    |
| 89  | 93  | 20 | Male   | 2nd Class | yes | 74 | Kochi   |
| 90  | 94  | 19 | Female | 3rd Class | yes | 53 | Gwalior |
| 91  | 95  | 21 | Male   | 3rd Class | yes | 50 | Bhopal  |
| 92  | 96  | 22 | Female | 3rd Class | yes | 40 | Chennai |
| 93  | 97  | 21 | Male   | 1st Class | no  | 87 | Delhi   |
| 94  | 98  | 20 | Female | 3rd Class | yes | 44 | Mumbai  |
| 95  | 99  | 20 | Male   | 1st Class | no  | 86 | Pune    |
| 96  | 100 | 19 | Male   | 1st Class | yes | 84 | Kochi   |
| 97  | 101 | 20 | Female | 3rd Class | no  | 43 | Gwalior |
| 98  | 102 | 19 | Male   | 1st Class | yes | 98 | Chennai |
| 99  | 103 | 19 | Female | 2nd Class | no  | 75 | Delhi   |
| 100 | 104 | 20 | Male   | 2nd Class | yes | 78 | Mumbai  |
| 101 | 105 | 18 | Female | 3rd Class | no  | 42 | Pune    |
| 102 | 106 | 21 | Male   | 2nd Class | yes | 77 | Kochi   |
| 103 | 107 | 19 | Male   | 2nd Class | no  | 78 | Gwalior |
| 104 | 108 | 20 | Female | 3rd Class | yes | 41 | Bhopal  |
| 105 | 109 | 19 | Male   | 2nd Class | no  | 79 | Chennai |
| 106 | 110 | 21 | Female | 2nd Class | yes | 76 | Delhi   |
| 107 | 111 | 22 | Male   | 3rd Class | no  | 45 | Mumbai  |
| 108 | 112 | 21 | Female | 1st Class | no  | 97 | Pune    |
| 109 | 113 | 20 | Male   | 2nd Class | no  | 75 | Kochi   |
| 110 | 114 | 20 | Male   | 3rd Class | no  | 46 | Gwalior |
| 111 | 115 | 19 | Female | 1st Class | no  | 96 | Bhopal  |
| 112 | 116 | 20 | Male   | 3rd Class | no  | 47 | Pune    |
| 113 | 117 | 19 | Male   | 1st Class | yes | 90 | Kochi   |
| 114 | 118 | 20 | Female | 2nd Class | no  | 79 | Gwalior |
| 115 | 119 | 18 | Male   | 1st Class | no  | 91 | Bhopal  |
| 116 | 120 | 21 | Female | 2nd Class | no  | 77 | Chennai |
| 117 | 121 | 19 | Male   | 1st Class | no  | 87 | Delhi   |
| 118 | 122 | 20 | Male   | 2nd Class | yes | 75 | Mumbai  |
| 119 | 123 | 19 | Female | 3rd Class | yes | 48 | Pune    |

|     |     |    |        |           |     |     |         |
|-----|-----|----|--------|-----------|-----|-----|---------|
| 120 | 124 | 21 | Male   | 3rd Class | yes | 49  | Chennai |
| 121 | 125 | 22 | Female | 3rd Class | yes | 55  | Delhi   |
| 122 | 126 | 21 | Male   | 1st Class | no  | 88  | Mumbai  |
| 123 | 127 | 20 | Female | 3rd Class | yes | 56  | Pune    |
| 124 | 128 | 20 | Male   | 1st Class | no  | 86  | Kochi   |
| 125 | 129 | 19 | Male   | 1st Class | yes | 89  | Gwalior |
| 126 | 130 | 20 | Female | 3rd Class | no  | 57  | Bhopal  |
| 127 | 131 | 19 | Male   | 1st Class | yes | 90  | Chennai |
| 128 | 132 | 19 | Female | 2nd Class | no  | 75  | Delhi   |
| 129 | 133 | 20 | Male   | 2nd Class | yes | 75  | Mumbai  |
| 130 | 134 | 18 | Female | 3rd Class | no  | 58  | Pune    |
| 131 | 135 | 21 | Male   | 2nd Class | yes | 75  | Kochi   |
| 132 | 136 | 19 | Male   | 2nd Class | no  | 74  | Gwalior |
| 133 | 137 | 20 | Female | 3rd Class | yes | 59  | Chennai |
| 134 | 138 | 19 | Male   | 2nd Class | no  | 73  | Delhi   |
| 135 | 139 | 21 | Female | 2nd Class | yes | 72  | Mumbai  |
| 136 | 140 | 22 | Male   | 3rd Class | no  | 50  | Pune    |
| 137 | 141 | 21 | Female | 1st Class | no  | 93  | Kochi   |
| 138 | 142 | 20 | Male   | 2nd Class | no  | 71  | Gwalior |
| 139 | 143 | 20 | Male   | 3rd Class | no  | 40  | Bhopal  |
| 140 | 144 | 19 | Female | 1st Class | no  | 97  | Chennai |
| 141 | 145 | 20 | Male   | 3rd Class | no  | 43  | Delhi   |
| 142 | 146 | 19 | Male   | 1st Class | yes | 98  | Mumbai  |
| 143 | 147 | 20 | Female | 2nd Class | no  | 70  | Pune    |
| 144 | 148 | 18 | Male   | 1st Class | no  | 99  | Kochi   |
| 145 | 149 | 21 | Female | 2nd Class | no  | 77  | Gwalior |
| 146 | 150 | 19 | Male   | 1st Class | no  | 100 | Bhopal  |
| 147 | 151 | 20 | Male   | 2nd Class | yes | 78  | Pune    |
| 148 | 152 | 19 | Female | 3rd Class | yes | 45  | Kochi   |
| 149 | 153 | 21 | Male   | 3rd Class | yes | 46  | Gwalior |
| 150 | 154 | 22 | Female | 3rd Class | yes | 47  | Bhopal  |
| 151 | 155 | 21 | Male   | 1st Class | no  | 89  | Chennai |
| 152 | 156 | 20 | Female | 3rd Class | yes | 49  | Delhi   |
| 153 | 157 | 20 | Male   | 1st Class | no  | 86  | Mumbai  |
| 154 | 158 | 19 | Male   | 1st Class | yes | 92  | Pune    |
| 155 | 159 | 20 | Female | 3rd Class | no  | 50  | Chennai |
| 156 | 160 | 19 | Male   | 1st Class | yes | 92  | Delhi   |
| 157 | 161 | 19 | Female | 2nd Class | no  | 79  | Mumbai  |
| 158 | 162 | 20 | Male   | 2nd Class | yes | 79  | Pune    |
| 159 | 163 | 18 | Female | 3rd Class | no  | 59  | Kochi   |
| 160 | 164 | 21 | Male   | 2nd Class | yes | 75  | Gwalior |
| 161 | 165 | 19 | Male   | 2nd Class | no  | 75  | Bhopal  |
| 162 | 166 | 20 | Female | 3rd Class | yes | 53  | Chennai |
| 163 | 167 | 19 | Male   | 2nd Class | no  | 74  | Delhi   |
| 164 | 168 | 21 | Female | 2nd Class | yes | 74  | Mumbai  |
| 165 | 169 | 22 | Male   | 3rd Class | no  | 53  | Pune    |
| 166 | 170 | 21 | Female | 1st Class | no  | 90  | Kochi   |
| 167 | 171 | 20 | Male   | 2nd Class | no  | 73  | Gwalior |
| 168 | 172 | 20 | Male   | 3rd Class | no  | 52  | Chennai |

|     |     |    |        |           |     |    |         |
|-----|-----|----|--------|-----------|-----|----|---------|
| 169 | 173 | 19 | Female | 1st Class | no  | 98 | Delhi   |
| 170 | 174 | 20 | Male   | 3rd Class | no  | 56 | Mumbai  |
| 171 | 175 | 19 | Male   | 1st Class | yes | 84 | Pune    |
| 172 | 176 | 20 | Female | 2nd Class | no  | 77 | Kochi   |
| 173 | 177 | 18 | Male   | 1st Class | no  | 97 | Gwalior |
| 174 | 178 | 21 | Female | 2nd Class | no  | 76 | Bhopal  |
| 175 | 179 | 19 | Male   | 1st Class | no  | 94 | Chennai |
| 176 | 180 | 20 | Male   | 2nd Class | yes | 76 | Delhi   |
| 177 | 181 | 19 | Female | 3rd Class | yes | 56 | Mumbai  |
| 178 | 182 | 21 | Male   | 3rd Class | yes | 58 | Pune    |
| 179 | 183 | 22 | Female | 3rd Class | yes | 59 | Kochi   |
| 180 | 184 | 21 | Male   | 1st Class | no  | 87 | Gwalior |
| 181 | 185 | 20 | Female | 3rd Class | yes | 50 | Bhopal  |
| 182 | 186 | 20 | Male   | 1st Class | no  | 85 | Pune    |
| 183 | 187 | 19 | Male   | 1st Class | yes | 83 | Kochi   |
| 184 | 188 | 20 | Female | 3rd Class | no  | 40 | Gwalior |
| 185 | 189 | 19 | Male   | 1st Class | yes | 89 | Bhopal  |
| 186 | 190 | 19 | Female | 2nd Class | no  | 75 | Chennai |
| 187 | 191 | 20 | Male   | 2nd Class | yes | 77 | Delhi   |
| 188 | 192 | 18 | Female | 3rd Class | no  | 44 | Mumbai  |
| 189 | 193 | 21 | Male   | 2nd Class | yes | 75 | Pune    |
| 190 | 194 | 19 | Male   | 2nd Class | no  | 74 | Chennai |
| 191 | 195 | 20 | Female | 3rd Class | yes | 43 | Delhi   |
| 192 | 196 | 19 | Male   | 2nd Class | no  | 74 | Mumbai  |
| 193 | 197 | 21 | Female | 2nd Class | yes | 74 | Pune    |
| 194 | 198 | 22 | Male   | 3rd Class | no  | 53 | Kochi   |
| 195 | 199 | 21 | Female | 1st Class | no  | 87 | Gwalior |
| 196 | 200 | 20 | Male   | 2nd Class | no  | 75 | Bhopal  |
| 197 | 201 | 20 | Male   | 3rd Class | no  | 42 | Chennai |
| 198 | 202 | 19 | Female | 1st Class | no  | 80 | Delhi   |
| 199 | 203 | 20 | Male   | 3rd Class | no  | 46 | Mumbai  |
| 200 | 204 | 19 | Male   | 1st Class | yes | 91 | Pune    |
| 201 | 205 | 20 | Female | 2nd Class | no  | 78 | Kochi   |
| 202 | 206 | 18 | Male   | 1st Class | no  | 80 | Gwalior |
| 203 | 207 | 21 | Female | 2nd Class | no  | 79 | Chennai |
| 204 | 208 | 19 | Male   | 1st Class | no  | 91 | Delhi   |
| 205 | 209 | 20 | Male   | 2nd Class | yes | 70 | Mumbai  |
| 206 | 210 | 19 | Female | 3rd Class | yes | 59 | Pune    |
| 207 | 211 | 21 | Male   | 3rd Class | yes | 58 | Kochi   |
| 208 | 212 | 22 | Female | 3rd Class | yes | 56 | Gwalior |
| 209 | 213 | 21 | Male   | 1st Class | no  | 92 | Bhopal  |
| 210 | 214 | 20 | Female | 3rd Class | yes | 54 | Chennai |
| 211 | 215 | 20 | Male   | 1st Class | no  | 93 | Delhi   |
| 212 | 216 | 19 | Male   | 1st Class | yes | 95 | Mumbai  |
| 213 | 217 | 20 | Female | 3rd Class | no  | 53 | Pune    |
| 214 | 218 | 19 | Male   | 1st Class | yes | 96 | Kochi   |
| 215 | 219 | 19 | Female | 2nd Class | no  | 75 | Gwalior |
| 216 | 220 | 20 | Male   | 2nd Class | yes | 75 | Bhopal  |
| 217 | 221 | 18 | Female | 3rd Class | no  | 52 | Pune    |

|     |     |    |        |           |     |    |         |
|-----|-----|----|--------|-----------|-----|----|---------|
| 218 | 222 | 21 | Male   | 2nd Class | yes | 74 | Kochi   |
| 219 | 223 | 19 | Male   | 2nd Class | no  | 79 | Gwalior |
| 220 | 224 | 20 | Female | 3rd Class | yes | 50 | Bhopal  |
| 221 | 225 | 19 | Male   | 2nd Class | no  | 72 | Chennai |
| 222 | 226 | 21 | Female | 2nd Class | yes | 71 | Delhi   |
| 223 | 227 | 22 | Male   | 3rd Class | no  | 51 | Mumbai  |
| 224 | 228 | 21 | Female | 1st Class | no  | 99 | Pune    |
| 225 | 229 | 20 | Male   | 2nd Class | no  | 70 | Chennai |
| 226 | 230 | 20 | Male   | 3rd Class | yes | 55 | Delhi   |
| 227 | 231 | 19 | Female | 1st Class | yes | 97 | Mumbai  |
| 228 | 232 | 20 | Male   | 3rd Class | yes | 59 | Pune    |

*##Q2.b) Max, Min marks gender wise*

```
max_marks_gender = combined_df.groupby("Gender")["Mark"].max()
```

```
print(max_marks_gender)
```

```
min_marks_gender = combined_df.groupby("Gender")["Mark"].min()
```

```
print(min_marks_gender)
```

Gender

Female      99

Male        100

Name: Mark, dtype: int64

Gender

Female      40

Male        40

Name: Mark, dtype: int64

*##Q2.c) students who scored more than average marks of class*

```
average_marks = combined_df['Mark'].mean()
```

```
above_average_students = combined_df[combined_df['Mark'] >
average_marks]
```

```
print(above_average_students[['Student_id', 'Mark']])
```

|    | Student_id | Mark |
|----|------------|------|
| 0  | 1          | 95   |
| 2  | 3          | 98   |
| 3  | 4          | 75   |
| 4  | 5          | 89   |
| 9  | 12         | 90   |
| 10 | 13         | 92   |
| 12 | 15         | 97   |
| 13 | 16         | 77   |
| 16 | 19         | 72   |
| 17 | 20         | 79   |
| 19 | 22         | 78   |
| 20 | 23         | 76   |
| 22 | 25         | 89   |
| 23 | 26         | 75   |
| 25 | 28         | 98   |

|     |     |    |
|-----|-----|----|
| 27  | 30  | 81 |
| 28  | 31  | 74 |
| 29  | 32  | 82 |
| 30  | 33  | 78 |
| 31  | 34  | 83 |
| 32  | 35  | 79 |
| 36  | 39  | 84 |
| 38  | 41  | 85 |
| 39  | 42  | 86 |
| 41  | 44  | 95 |
| 42  | 45  | 73 |
| 43  | 46  | 79 |
| 50  | 54  | 82 |
| 53  | 57  | 85 |
| 55  | 59  | 89 |
| 57  | 61  | 98 |
| 58  | 62  | 77 |
| 59  | 63  | 98 |
| 60  | 64  | 78 |
| 64  | 68  | 87 |
| 66  | 70  | 86 |
| 67  | 71  | 85 |
| 69  | 73  | 89 |
| 70  | 74  | 77 |
| 71  | 75  | 79 |
| 73  | 77  | 77 |
| 74  | 78  | 76 |
| 76  | 80  | 75 |
| 77  | 81  | 74 |
| 79  | 83  | 98 |
| 80  | 84  | 73 |
| 82  | 86  | 97 |
| 84  | 88  | 96 |
| 85  | 89  | 75 |
| 86  | 90  | 94 |
| 87  | 91  | 74 |
| 88  | 92  | 93 |
| 89  | 93  | 74 |
| 93  | 97  | 87 |
| 95  | 99  | 86 |
| 96  | 100 | 84 |
| 98  | 102 | 98 |
| 99  | 103 | 75 |
| 100 | 104 | 78 |
| 102 | 106 | 77 |
| 103 | 107 | 78 |
| 105 | 109 | 79 |
| 106 | 110 | 76 |
| 108 | 112 | 97 |



|     |     |     |
|-----|-----|-----|
| 109 | 113 | 75  |
| 111 | 115 | 96  |
| 113 | 117 | 90  |
| 114 | 118 | 79  |
| 115 | 119 | 91  |
| 116 | 120 | 77  |
| 117 | 121 | 87  |
| 118 | 122 | 75  |
| 122 | 126 | 88  |
| 124 | 128 | 86  |
| 125 | 129 | 89  |
| 127 | 131 | 90  |
| 128 | 132 | 75  |
| 129 | 133 | 75  |
| 131 | 135 | 75  |
| 132 | 136 | 74  |
| 134 | 138 | 73  |
| 135 | 139 | 72  |
| 137 | 141 | 93  |
| 140 | 144 | 97  |
| 142 | 146 | 98  |
| 144 | 148 | 99  |
| 145 | 149 | 77  |
| 146 | 150 | 100 |
| 147 | 151 | 78  |
| 151 | 155 | 89  |
| 153 | 157 | 86  |
| 154 | 158 | 92  |
| 156 | 160 | 92  |
| 157 | 161 | 79  |
| 158 | 162 | 79  |
| 160 | 164 | 75  |
| 161 | 165 | 75  |
| 163 | 167 | 74  |
| 164 | 168 | 74  |
| 166 | 170 | 90  |
| 167 | 171 | 73  |
| 169 | 173 | 98  |
| 171 | 175 | 84  |
| 172 | 176 | 77  |
| 173 | 177 | 97  |
| 174 | 178 | 76  |
| 175 | 179 | 94  |
| 176 | 180 | 76  |
| 180 | 184 | 87  |
| 182 | 186 | 85  |
| 183 | 187 | 83  |
| 185 | 189 | 89  |
| 186 | 190 | 75  |

|     |     |    |
|-----|-----|----|
| 187 | 191 | 77 |
| 189 | 193 | 75 |
| 190 | 194 | 74 |
| 192 | 196 | 74 |
| 193 | 197 | 74 |
| 195 | 199 | 87 |
| 196 | 200 | 75 |
| 198 | 202 | 80 |
| 200 | 204 | 91 |
| 201 | 205 | 78 |
| 202 | 206 | 80 |
| 203 | 207 | 79 |
| 204 | 208 | 91 |
| 209 | 213 | 92 |
| 211 | 215 | 93 |
| 212 | 216 | 95 |
| 214 | 218 | 96 |
| 215 | 219 | 75 |
| 216 | 220 | 75 |
| 218 | 222 | 74 |
| 219 | 223 | 79 |
| 221 | 225 | 72 |
| 224 | 228 | 99 |
| 227 | 231 | 97 |

*##Q2.d) males who are employed*

```
male_and_employed = combined_df[(combined_df['Gender'] == 'Male') &
(combined_df['Employed'] == 'yes')]
print(male_and_employed)
```

|    | Student_id | Age | Gender | Grade     | Employed | Mark | City    |
|----|------------|-----|--------|-----------|----------|------|---------|
| 0  | 1          | 19  | Male   | 1st Class | yes      | 95   | Chennai |
| 5  | 6          | 20  | Male   | 2nd Class | yes      | 69   | Gwalior |
| 7  | 8          | 21  | Male   | 3rd Class | yes      | 54   | Chennai |
| 10 | 13         | 19  | Male   | 1st Class | yes      | 92   | Gwalior |
| 12 | 15         | 19  | Male   | 1st Class | yes      | 97   | Pune    |
| 14 | 17         | 20  | Male   | 2nd Class | yes      | 68   | Gwalior |
| 16 | 19         | 21  | Male   | 2nd Class | yes      | 72   | Chennai |
| 27 | 30         | 19  | Male   | 1st Class | yes      | 81   | Chennai |
| 32 | 35         | 20  | Male   | 2nd Class | yes      | 79   | Gwalior |
| 34 | 37         | 21  | Male   | 3rd Class | yes      | 53   | Pune    |
| 39 | 42         | 19  | Male   | 1st Class | yes      | 86   | Delhi   |
| 41 | 44         | 19  | Male   | 1st Class | yes      | 95   | Pune    |
| 43 | 46         | 20  | Male   | 2nd Class | yes      | 79   | Delhi   |
| 45 | 48         | 21  | Male   | 2nd Class | yes      | 67   | Pune    |
| 55 | 59         | 19  | Male   | 1st Class | yes      | 89   | Pune    |
| 60 | 64         | 20  | Male   | 2nd Class | yes      | 78   | Delhi   |
| 62 | 66         | 21  | Male   | 3rd Class | yes      | 59   | Pune    |
| 67 | 71         | 19  | Male   | 1st Class | yes      | 85   | Kochi   |
| 69 | 73         | 19  | Male   | 1st Class | yes      | 89   | Bhopal  |

|     |     |    |      |           |     |    |         |
|-----|-----|----|------|-----------|-----|----|---------|
| 71  | 75  | 20 | Male | 2nd Class | yes | 79 | Delhi   |
| 73  | 77  | 21 | Male | 2nd Class | yes | 77 | Pune    |
| 84  | 88  | 19 | Male | 1st Class | yes | 96 | Pune    |
| 89  | 93  | 20 | Male | 2nd Class | yes | 74 | Kochi   |
| 91  | 95  | 21 | Male | 3rd Class | yes | 50 | Bhopal  |
| 96  | 100 | 19 | Male | 1st Class | yes | 84 | Kochi   |
| 98  | 102 | 19 | Male | 1st Class | yes | 98 | Chennai |
| 100 | 104 | 20 | Male | 2nd Class | yes | 78 | Mumbai  |
| 102 | 106 | 21 | Male | 2nd Class | yes | 77 | Kochi   |
| 113 | 117 | 19 | Male | 1st Class | yes | 90 | Kochi   |
| 118 | 122 | 20 | Male | 2nd Class | yes | 75 | Mumbai  |
| 120 | 124 | 21 | Male | 3rd Class | yes | 49 | Chennai |
| 125 | 129 | 19 | Male | 1st Class | yes | 89 | Gwalior |
| 127 | 131 | 19 | Male | 1st Class | yes | 90 | Chennai |
| 129 | 133 | 20 | Male | 2nd Class | yes | 75 | Mumbai  |
| 131 | 135 | 21 | Male | 2nd Class | yes | 75 | Kochi   |
| 142 | 146 | 19 | Male | 1st Class | yes | 98 | Mumbai  |
| 147 | 151 | 20 | Male | 2nd Class | yes | 78 | Pune    |
| 149 | 153 | 21 | Male | 3rd Class | yes | 46 | Gwalior |
| 154 | 158 | 19 | Male | 1st Class | yes | 92 | Pune    |
| 156 | 160 | 19 | Male | 1st Class | yes | 92 | Delhi   |
| 158 | 162 | 20 | Male | 2nd Class | yes | 79 | Pune    |
| 160 | 164 | 21 | Male | 2nd Class | yes | 75 | Gwalior |
| 171 | 175 | 19 | Male | 1st Class | yes | 84 | Pune    |
| 176 | 180 | 20 | Male | 2nd Class | yes | 76 | Delhi   |
| 178 | 182 | 21 | Male | 3rd Class | yes | 58 | Pune    |
| 183 | 187 | 19 | Male | 1st Class | yes | 83 | Kochi   |
| 185 | 189 | 19 | Male | 1st Class | yes | 89 | Bhopal  |
| 187 | 191 | 20 | Male | 2nd Class | yes | 77 | Delhi   |
| 189 | 193 | 21 | Male | 2nd Class | yes | 75 | Pune    |
| 200 | 204 | 19 | Male | 1st Class | yes | 91 | Pune    |
| 205 | 209 | 20 | Male | 2nd Class | yes | 70 | Mumbai  |
| 207 | 211 | 21 | Male | 3rd Class | yes | 58 | Kochi   |
| 212 | 216 | 19 | Male | 1st Class | yes | 95 | Mumbai  |
| 214 | 218 | 19 | Male | 1st Class | yes | 96 | Kochi   |
| 216 | 220 | 20 | Male | 2nd Class | yes | 75 | Bhopal  |
| 218 | 222 | 21 | Male | 2nd Class | yes | 74 | Kochi   |
| 226 | 230 | 20 | Male | 3rd Class | yes | 55 | Delhi   |
| 228 | 232 | 20 | Male | 3rd Class | yes | 59 | Pune    |

*##Q2.e) Include IQ\_level field based on secured marks range*

```
def classify_iq(Mark):
    if Mark > 80:
        return 'Intelligent'
    elif 50 <= Mark <= 80:
        return 'Mediocre'
    else:
        return 'Weak'
combined_df['IQ_level'] = combined_df['Mark'].apply(classify_iq)
print(combined_df)
```

|             | Student_id | Age | Gender | Grade     | Employed | Mark | City    |
|-------------|------------|-----|--------|-----------|----------|------|---------|
| IQ_level    |            |     |        |           |          |      |         |
| 0           | 1          | 19  | Male   | 1st Class | yes      | 95   | Chennai |
| Intelligent |            |     |        |           |          |      |         |
| 1           | 2          | 20  | Female | 2nd Class | no       | 70   | Delhi   |
| Mediocre    |            |     |        |           |          |      |         |
| 2           | 3          | 18  | Male   | 1st Class | no       | 98   | Mumbai  |
| Intelligent |            |     |        |           |          |      |         |
| 3           | 4          | 21  | Female | 2nd Class | no       | 75   | Pune    |
| Mediocre    |            |     |        |           |          |      |         |
| 4           | 5          | 19  | Male   | 1st Class | no       | 89   | Kochi   |
| Intelligent |            |     |        |           |          |      |         |
| 5           | 6          | 20  | Male   | 2nd Class | yes      | 69   | Gwalior |
| Mediocre    |            |     |        |           |          |      |         |
| 6           | 7          | 19  | Female | 3rd Class | yes      | 52   | Bhopal  |
| Mediocre    |            |     |        |           |          |      |         |
| 7           | 8          | 21  | Male   | 3rd Class | yes      | 54   | Chennai |
| Mediocre    |            |     |        |           |          |      |         |
| 8           | 11         | 20  | Female | 3rd Class | yes      | 52   | Pune    |
| Mediocre    |            |     |        |           |          |      |         |
| 9           | 12         | 20  | Male   | 1st Class | no       | 90   | Kochi   |
| Intelligent |            |     |        |           |          |      |         |
| 10          | 13         | 19  | Male   | 1st Class | yes      | 92   | Gwalior |
| Intelligent |            |     |        |           |          |      |         |
| 11          | 14         | 20  | Female | 3rd Class | no       | 55   | Bhopal  |
| Mediocre    |            |     |        |           |          |      |         |
| 12          | 15         | 19  | Male   | 1st Class | yes      | 97   | Pune    |
| Intelligent |            |     |        |           |          |      |         |
| 13          | 16         | 19  | Female | 2nd Class | no       | 77   | Kochi   |
| Mediocre    |            |     |        |           |          |      |         |
| 14          | 17         | 20  | Male   | 2nd Class | yes      | 68   | Gwalior |
| Mediocre    |            |     |        |           |          |      |         |
| 15          | 18         | 18  | Female | 3rd Class | no       | 51   | Bhopal  |
| Mediocre    |            |     |        |           |          |      |         |
| 16          | 19         | 21  | Male   | 2nd Class | yes      | 72   | Chennai |
| Mediocre    |            |     |        |           |          |      |         |
| 17          | 20         | 19  | Male   | 2nd Class | no       | 79   | Delhi   |
| Mediocre    |            |     |        |           |          |      |         |
| 18          | 21         | 20  | Female | 3rd Class | yes      | 49   | Mumbai  |
| Weak        |            |     |        |           |          |      |         |
| 19          | 22         | 19  | Male   | 2nd Class | no       | 78   | Pune    |
| Mediocre    |            |     |        |           |          |      |         |
| 20          | 23         | 21  | Female | 2nd Class | yes      | 76   | Chennai |
| Mediocre    |            |     |        |           |          |      |         |
| 21          | 24         | 22  | Male   | 3rd Class | no       | 43   | Delhi   |
| Weak        |            |     |        |           |          |      |         |
| 22          | 25         | 21  | Female | 1st Class | no       | 89   | Mumbai  |
| Intelligent |            |     |        |           |          |      |         |
| 23          | 26         | 20  | Male   | 2nd Class | no       | 75   | Pune    |
| Mediocre    |            |     |        |           |          |      |         |

|             |    |    |        |           |     |    |         |
|-------------|----|----|--------|-----------|-----|----|---------|
| 24          | 27 | 20 | Male   | 3rd Class | no  | 45 | Kochi   |
| Weak        |    |    |        |           |     |    |         |
| 25          | 28 | 19 | Female | 1st Class | no  | 98 | Gwalior |
| Intelligent |    |    |        |           |     |    |         |
| 26          | 29 | 20 | Male   | 3rd Class | no  | 54 | Bhopal  |
| Mediocre    |    |    |        |           |     |    |         |
| 27          | 30 | 19 | Male   | 1st Class | yes | 81 | Chennai |
| Intelligent |    |    |        |           |     |    |         |
| 28          | 31 | 20 | Female | 2nd Class | no  | 74 | Delhi   |
| Mediocre    |    |    |        |           |     |    |         |
| 29          | 32 | 18 | Male   | 1st Class | no  | 82 | Mumbai  |
| Intelligent |    |    |        |           |     |    |         |
| 30          | 33 | 21 | Female | 2nd Class | no  | 78 | Pune    |
| Mediocre    |    |    |        |           |     |    |         |
| 31          | 34 | 19 | Male   | 1st Class | no  | 83 | Kochi   |
| Intelligent |    |    |        |           |     |    |         |
| 32          | 35 | 20 | Male   | 2nd Class | yes | 79 | Gwalior |
| Mediocre    |    |    |        |           |     |    |         |
| 33          | 36 | 19 | Female | 3rd Class | yes | 55 | Bhopal  |
| Mediocre    |    |    |        |           |     |    |         |
| 34          | 37 | 21 | Male   | 3rd Class | yes | 53 | Pune    |
| Mediocre    |    |    |        |           |     |    |         |
| 35          | 38 | 22 | Female | 3rd Class | yes | 55 | Kochi   |
| Mediocre    |    |    |        |           |     |    |         |
| 36          | 39 | 21 | Male   | 1st Class | no  | 84 | Gwalior |
| Intelligent |    |    |        |           |     |    |         |
| 37          | 40 | 20 | Female | 3rd Class | yes | 44 | Bhopal  |
| Weak        |    |    |        |           |     |    |         |
| 38          | 41 | 20 | Male   | 1st Class | no  | 85 | Chennai |
| Intelligent |    |    |        |           |     |    |         |
| 39          | 42 | 19 | Male   | 1st Class | yes | 86 | Delhi   |
| Intelligent |    |    |        |           |     |    |         |
| 40          | 43 | 20 | Female | 3rd Class | no  | 45 | Mumbai  |
| Weak        |    |    |        |           |     |    |         |
| 41          | 44 | 19 | Male   | 1st Class | yes | 95 | Pune    |
| Intelligent |    |    |        |           |     |    |         |
| 42          | 45 | 19 | Female | 2nd Class | no  | 73 | Chennai |
| Mediocre    |    |    |        |           |     |    |         |
| 43          | 46 | 20 | Male   | 2nd Class | yes | 79 | Delhi   |
| Mediocre    |    |    |        |           |     |    |         |
| 44          | 47 | 18 | Female | 3rd Class | no  | 46 | Mumbai  |
| Weak        |    |    |        |           |     |    |         |
| 45          | 48 | 21 | Male   | 2nd Class | yes | 67 | Pune    |
| Mediocre    |    |    |        |           |     |    |         |
| 46          | 50 | 20 | Female | 3rd Class | yes | 47 | Gwalior |
| Weak        |    |    |        |           |     |    |         |
| 47          | 51 | 19 | Male   | 2nd Class | no  | 65 | Bhopal  |
| Mediocre    |    |    |        |           |     |    |         |
| 48          | 52 | 21 | Female | 2nd Class | yes | 67 | Chennai |
| Mediocre    |    |    |        |           |     |    |         |

|             |    |    |        |           |     |    |         |
|-------------|----|----|--------|-----------|-----|----|---------|
| 49          | 53 | 22 | Male   | 3rd Class | no  | 49 | Delhi   |
| Weak        |    |    |        |           |     |    |         |
| 50          | 54 | 21 | Female | 1st Class | no  | 82 | Mumbai  |
| Intelligent |    |    |        |           |     |    |         |
| 51          | 55 | 20 | Male   | 2nd Class | no  | 69 | Pune    |
| Mediocre    |    |    |        |           |     |    |         |
| 52          | 56 | 20 | Male   | 3rd Class | no  | 48 | Kochi   |
| Weak        |    |    |        |           |     |    |         |
| 53          | 57 | 19 | Female | 1st Class | no  | 85 | Gwalior |
| Intelligent |    |    |        |           |     |    |         |
| 54          | 58 | 20 | Male   | 3rd Class | no  | 42 | Bhopal  |
| Weak        |    |    |        |           |     |    |         |
| 55          | 59 | 19 | Male   | 1st Class | yes | 89 | Pune    |
| Intelligent |    |    |        |           |     |    |         |
| 56          | 60 | 20 | Female | 2nd Class | no  | 66 | Kochi   |
| Mediocre    |    |    |        |           |     |    |         |
| 57          | 61 | 18 | Male   | 1st Class | no  | 98 | Gwalior |
| Intelligent |    |    |        |           |     |    |         |
| 58          | 62 | 21 | Female | 2nd Class | no  | 77 | Bhopal  |
| Mediocre    |    |    |        |           |     |    |         |
| 59          | 63 | 19 | Male   | 1st Class | no  | 98 | Chennai |
| Intelligent |    |    |        |           |     |    |         |
| 60          | 64 | 20 | Male   | 2nd Class | yes | 78 | Delhi   |
| Mediocre    |    |    |        |           |     |    |         |
| 61          | 65 | 19 | Female | 3rd Class | yes | 43 | Mumbai  |
| Weak        |    |    |        |           |     |    |         |
| 62          | 66 | 21 | Male   | 3rd Class | yes | 59 | Pune    |
| Mediocre    |    |    |        |           |     |    |         |
| 63          | 67 | 22 | Female | 3rd Class | yes | 58 | Chennai |
| Mediocre    |    |    |        |           |     |    |         |
| 64          | 68 | 21 | Male   | 1st Class | no  | 87 | Delhi   |
| Intelligent |    |    |        |           |     |    |         |
| 65          | 69 | 20 | Female | 3rd Class | yes | 55 | Mumbai  |
| Mediocre    |    |    |        |           |     |    |         |
| 66          | 70 | 20 | Male   | 1st Class | no  | 86 | Pune    |
| Intelligent |    |    |        |           |     |    |         |
| 67          | 71 | 19 | Male   | 1st Class | yes | 85 | Kochi   |
| Intelligent |    |    |        |           |     |    |         |
| 68          | 72 | 20 | Female | 3rd Class | no  | 54 | Gwalior |
| Mediocre    |    |    |        |           |     |    |         |
| 69          | 73 | 19 | Male   | 1st Class | yes | 89 | Bhopal  |
| Intelligent |    |    |        |           |     |    |         |
| 70          | 74 | 19 | Female | 2nd Class | no  | 77 | Chennai |
| Mediocre    |    |    |        |           |     |    |         |
| 71          | 75 | 20 | Male   | 2nd Class | yes | 79 | Delhi   |
| Mediocre    |    |    |        |           |     |    |         |
| 72          | 76 | 18 | Female | 3rd Class | no  | 53 | Mumbai  |
| Mediocre    |    |    |        |           |     |    |         |
| 73          | 77 | 21 | Male   | 2nd Class | yes | 77 | Pune    |
| Mediocre    |    |    |        |           |     |    |         |

|                   |     |    |        |           |     |    |         |
|-------------------|-----|----|--------|-----------|-----|----|---------|
| 74<br>Mediocre    | 78  | 19 | Male   | 2nd Class | no  | 76 | Kochi   |
| 75<br>Mediocre    | 79  | 20 | Female | 3rd Class | yes | 56 | Gwalior |
| 76<br>Mediocre    | 80  | 19 | Male   | 2nd Class | no  | 75 | Bhopal  |
| 77<br>Mediocre    | 81  | 21 | Female | 2nd Class | yes | 74 | Pune    |
| 78<br>Mediocre    | 82  | 22 | Male   | 3rd Class | no  | 59 | Kochi   |
| 79<br>Intelligent | 83  | 21 | Female | 1st Class | no  | 98 | Gwalior |
| 80<br>Mediocre    | 84  | 20 | Male   | 2nd Class | no  | 73 | Bhopal  |
| 81<br>Mediocre    | 85  | 20 | Male   | 3rd Class | no  | 58 | Chennai |
| 82<br>Intelligent | 86  | 19 | Female | 1st Class | no  | 97 | Delhi   |
| 83<br>Mediocre    | 87  | 20 | Male   | 3rd Class | no  | 57 | Mumbai  |
| 84<br>Intelligent | 88  | 19 | Male   | 1st Class | yes | 96 | Pune    |
| 85<br>Mediocre    | 89  | 20 | Female | 2nd Class | no  | 75 | Chennai |
| 86<br>Intelligent | 90  | 18 | Male   | 1st Class | no  | 94 | Delhi   |
| 87<br>Mediocre    | 91  | 21 | Female | 2nd Class | no  | 74 | Mumbai  |
| 88<br>Intelligent | 92  | 19 | Male   | 1st Class | no  | 93 | Pune    |
| 89<br>Mediocre    | 93  | 20 | Male   | 2nd Class | yes | 74 | Kochi   |
| 90<br>Mediocre    | 94  | 19 | Female | 3rd Class | yes | 53 | Gwalior |
| 91<br>Mediocre    | 95  | 21 | Male   | 3rd Class | yes | 50 | Bhopal  |
| 92<br>Weak        | 96  | 22 | Female | 3rd Class | yes | 40 | Chennai |
| 93<br>Intelligent | 97  | 21 | Male   | 1st Class | no  | 87 | Delhi   |
| 94<br>Weak        | 98  | 20 | Female | 3rd Class | yes | 44 | Mumbai  |
| 95<br>Intelligent | 99  | 20 | Male   | 1st Class | no  | 86 | Pune    |
| 96<br>Intelligent | 100 | 19 | Male   | 1st Class | yes | 84 | Kochi   |
| 97<br>Weak        | 101 | 20 | Female | 3rd Class | no  | 43 | Gwalior |
| 98<br>Intelligent | 102 | 19 | Male   | 1st Class | yes | 98 | Chennai |

|             |     |    |        |           |     |    |         |
|-------------|-----|----|--------|-----------|-----|----|---------|
| 99          | 103 | 19 | Female | 2nd Class | no  | 75 | Delhi   |
| Mediocre    |     |    |        |           |     |    |         |
| 100         | 104 | 20 | Male   | 2nd Class | yes | 78 | Mumbai  |
| Mediocre    |     |    |        |           |     |    |         |
| 101         | 105 | 18 | Female | 3rd Class | no  | 42 | Pune    |
| Weak        |     |    |        |           |     |    |         |
| 102         | 106 | 21 | Male   | 2nd Class | yes | 77 | Kochi   |
| Mediocre    |     |    |        |           |     |    |         |
| 103         | 107 | 19 | Male   | 2nd Class | no  | 78 | Gwalior |
| Mediocre    |     |    |        |           |     |    |         |
| 104         | 108 | 20 | Female | 3rd Class | yes | 41 | Bhopal  |
| Weak        |     |    |        |           |     |    |         |
| 105         | 109 | 19 | Male   | 2nd Class | no  | 79 | Chennai |
| Mediocre    |     |    |        |           |     |    |         |
| 106         | 110 | 21 | Female | 2nd Class | yes | 76 | Delhi   |
| Mediocre    |     |    |        |           |     |    |         |
| 107         | 111 | 22 | Male   | 3rd Class | no  | 45 | Mumbai  |
| Weak        |     |    |        |           |     |    |         |
| 108         | 112 | 21 | Female | 1st Class | no  | 97 | Pune    |
| Intelligent |     |    |        |           |     |    |         |
| 109         | 113 | 20 | Male   | 2nd Class | no  | 75 | Kochi   |
| Mediocre    |     |    |        |           |     |    |         |
| 110         | 114 | 20 | Male   | 3rd Class | no  | 46 | Gwalior |
| Weak        |     |    |        |           |     |    |         |
| 111         | 115 | 19 | Female | 1st Class | no  | 96 | Bhopal  |
| Intelligent |     |    |        |           |     |    |         |
| 112         | 116 | 20 | Male   | 3rd Class | no  | 47 | Pune    |
| Weak        |     |    |        |           |     |    |         |
| 113         | 117 | 19 | Male   | 1st Class | yes | 90 | Kochi   |
| Intelligent |     |    |        |           |     |    |         |
| 114         | 118 | 20 | Female | 2nd Class | no  | 79 | Gwalior |
| Mediocre    |     |    |        |           |     |    |         |
| 115         | 119 | 18 | Male   | 1st Class | no  | 91 | Bhopal  |
| Intelligent |     |    |        |           |     |    |         |
| 116         | 120 | 21 | Female | 2nd Class | no  | 77 | Chennai |
| Mediocre    |     |    |        |           |     |    |         |
| 117         | 121 | 19 | Male   | 1st Class | no  | 87 | Delhi   |
| Intelligent |     |    |        |           |     |    |         |
| 118         | 122 | 20 | Male   | 2nd Class | yes | 75 | Mumbai  |
| Mediocre    |     |    |        |           |     |    |         |
| 119         | 123 | 19 | Female | 3rd Class | yes | 48 | Pune    |
| Weak        |     |    |        |           |     |    |         |
| 120         | 124 | 21 | Male   | 3rd Class | yes | 49 | Chennai |
| Weak        |     |    |        |           |     |    |         |
| 121         | 125 | 22 | Female | 3rd Class | yes | 55 | Delhi   |
| Mediocre    |     |    |        |           |     |    |         |
| 122         | 126 | 21 | Male   | 1st Class | no  | 88 | Mumbai  |
| Intelligent |     |    |        |           |     |    |         |
| 123         | 127 | 20 | Female | 3rd Class | yes | 56 | Pune    |
| Mediocre    |     |    |        |           |     |    |         |



|             |     |    |        |           |     |     |         |
|-------------|-----|----|--------|-----------|-----|-----|---------|
| 124         | 128 | 20 | Male   | 1st Class | no  | 86  | Kochi   |
| Intelligent | 125 | 19 | Male   | 1st Class | yes | 89  | Gwalior |
| Intelligent | 126 | 20 | Female | 3rd Class | no  | 57  | Bhopal  |
| Mediocre    | 127 | 19 | Male   | 1st Class | yes | 90  | Chennai |
| Intelligent | 128 | 19 | Female | 2nd Class | no  | 75  | Delhi   |
| Mediocre    | 129 | 20 | Male   | 2nd Class | yes | 75  | Mumbai  |
| Mediocre    | 130 | 18 | Female | 3rd Class | no  | 58  | Pune    |
| Mediocre    | 131 | 21 | Male   | 2nd Class | yes | 75  | Kochi   |
| Mediocre    | 132 | 19 | Male   | 2nd Class | no  | 74  | Gwalior |
| Mediocre    | 133 | 20 | Female | 3rd Class | yes | 59  | Chennai |
| Mediocre    | 134 | 19 | Male   | 2nd Class | no  | 73  | Delhi   |
| Mediocre    | 135 | 21 | Female | 2nd Class | yes | 72  | Mumbai  |
| Mediocre    | 136 | 22 | Male   | 3rd Class | no  | 50  | Pune    |
| Mediocre    | 137 | 21 | Female | 1st Class | no  | 93  | Kochi   |
| Intelligent | 138 | 20 | Male   | 2nd Class | no  | 71  | Gwalior |
| Mediocre    | 139 | 20 | Male   | 3rd Class | no  | 40  | Bhopal  |
| Weak        | 140 | 19 | Female | 1st Class | no  | 97  | Chennai |
| Intelligent | 141 | 20 | Male   | 3rd Class | no  | 43  | Delhi   |
| Weak        | 142 | 19 | Male   | 1st Class | yes | 98  | Mumbai  |
| Intelligent | 143 | 20 | Female | 2nd Class | no  | 70  | Pune    |
| Mediocre    | 144 | 18 | Male   | 1st Class | no  | 99  | Kochi   |
| Intelligent | 145 | 21 | Female | 2nd Class | no  | 77  | Gwalior |
| Mediocre    | 146 | 19 | Male   | 1st Class | no  | 100 | Bhopal  |
| Intelligent | 147 | 20 | Male   | 2nd Class | yes | 78  | Pune    |
| Mediocre    | 148 | 19 | Female | 3rd Class | yes | 45  | Kochi   |
| Weak        |     |    |        |           |     |     |         |

|             |     |    |        |           |     |    |         |
|-------------|-----|----|--------|-----------|-----|----|---------|
| 149         | 153 | 21 | Male   | 3rd Class | yes | 46 | Gwalior |
| Weak        |     |    |        |           |     |    |         |
| 150         | 154 | 22 | Female | 3rd Class | yes | 47 | Bhopal  |
| Weak        |     |    |        |           |     |    |         |
| 151         | 155 | 21 | Male   | 1st Class | no  | 89 | Chennai |
| Intelligent |     |    |        |           |     |    |         |
| 152         | 156 | 20 | Female | 3rd Class | yes | 49 | Delhi   |
| Weak        |     |    |        |           |     |    |         |
| 153         | 157 | 20 | Male   | 1st Class | no  | 86 | Mumbai  |
| Intelligent |     |    |        |           |     |    |         |
| 154         | 158 | 19 | Male   | 1st Class | yes | 92 | Pune    |
| Intelligent |     |    |        |           |     |    |         |
| 155         | 159 | 20 | Female | 3rd Class | no  | 50 | Chennai |
| Mediocre    |     |    |        |           |     |    |         |
| 156         | 160 | 19 | Male   | 1st Class | yes | 92 | Delhi   |
| Intelligent |     |    |        |           |     |    |         |
| 157         | 161 | 19 | Female | 2nd Class | no  | 79 | Mumbai  |
| Mediocre    |     |    |        |           |     |    |         |
| 158         | 162 | 20 | Male   | 2nd Class | yes | 79 | Pune    |
| Mediocre    |     |    |        |           |     |    |         |
| 159         | 163 | 18 | Female | 3rd Class | no  | 59 | Kochi   |
| Mediocre    |     |    |        |           |     |    |         |
| 160         | 164 | 21 | Male   | 2nd Class | yes | 75 | Gwalior |
| Mediocre    |     |    |        |           |     |    |         |
| 161         | 165 | 19 | Male   | 2nd Class | no  | 75 | Bhopal  |
| Mediocre    |     |    |        |           |     |    |         |
| 162         | 166 | 20 | Female | 3rd Class | yes | 53 | Chennai |
| Mediocre    |     |    |        |           |     |    |         |
| 163         | 167 | 19 | Male   | 2nd Class | no  | 74 | Delhi   |
| Mediocre    |     |    |        |           |     |    |         |
| 164         | 168 | 21 | Female | 2nd Class | yes | 74 | Mumbai  |
| Mediocre    |     |    |        |           |     |    |         |
| 165         | 169 | 22 | Male   | 3rd Class | no  | 53 | Pune    |
| Mediocre    |     |    |        |           |     |    |         |
| 166         | 170 | 21 | Female | 1st Class | no  | 90 | Kochi   |
| Intelligent |     |    |        |           |     |    |         |
| 167         | 171 | 20 | Male   | 2nd Class | no  | 73 | Gwalior |
| Mediocre    |     |    |        |           |     |    |         |
| 168         | 172 | 20 | Male   | 3rd Class | no  | 52 | Chennai |
| Mediocre    |     |    |        |           |     |    |         |
| 169         | 173 | 19 | Female | 1st Class | no  | 98 | Delhi   |
| Intelligent |     |    |        |           |     |    |         |
| 170         | 174 | 20 | Male   | 3rd Class | no  | 56 | Mumbai  |
| Mediocre    |     |    |        |           |     |    |         |
| 171         | 175 | 19 | Male   | 1st Class | yes | 84 | Pune    |
| Intelligent |     |    |        |           |     |    |         |
| 172         | 176 | 20 | Female | 2nd Class | no  | 77 | Kochi   |
| Mediocre    |     |    |        |           |     |    |         |
| 173         | 177 | 18 | Male   | 1st Class | no  | 97 | Gwalior |
| Intelligent |     |    |        |           |     |    |         |

|             |     |    |        |           |     |    |         |
|-------------|-----|----|--------|-----------|-----|----|---------|
| 174         | 178 | 21 | Female | 2nd Class | no  | 76 | Bhopal  |
| Mediocre    |     |    |        |           |     |    |         |
| 175         | 179 | 19 | Male   | 1st Class | no  | 94 | Chennai |
| Intelligent |     |    |        |           |     |    |         |
| 176         | 180 | 20 | Male   | 2nd Class | yes | 76 | Delhi   |
| Mediocre    |     |    |        |           |     |    |         |
| 177         | 181 | 19 | Female | 3rd Class | yes | 56 | Mumbai  |
| Mediocre    |     |    |        |           |     |    |         |
| 178         | 182 | 21 | Male   | 3rd Class | yes | 58 | Pune    |
| Mediocre    |     |    |        |           |     |    |         |
| 179         | 183 | 22 | Female | 3rd Class | yes | 59 | Kochi   |
| Mediocre    |     |    |        |           |     |    |         |
| 180         | 184 | 21 | Male   | 1st Class | no  | 87 | Gwalior |
| Intelligent |     |    |        |           |     |    |         |
| 181         | 185 | 20 | Female | 3rd Class | yes | 50 | Bhopal  |
| Mediocre    |     |    |        |           |     |    |         |
| 182         | 186 | 20 | Male   | 1st Class | no  | 85 | Pune    |
| Intelligent |     |    |        |           |     |    |         |
| 183         | 187 | 19 | Male   | 1st Class | yes | 83 | Kochi   |
| Intelligent |     |    |        |           |     |    |         |
| 184         | 188 | 20 | Female | 3rd Class | no  | 40 | Gwalior |
| Weak        |     |    |        |           |     |    |         |
| 185         | 189 | 19 | Male   | 1st Class | yes | 89 | Bhopal  |
| Intelligent |     |    |        |           |     |    |         |
| 186         | 190 | 19 | Female | 2nd Class | no  | 75 | Chennai |
| Mediocre    |     |    |        |           |     |    |         |
| 187         | 191 | 20 | Male   | 2nd Class | yes | 77 | Delhi   |
| Mediocre    |     |    |        |           |     |    |         |
| 188         | 192 | 18 | Female | 3rd Class | no  | 44 | Mumbai  |
| Weak        |     |    |        |           |     |    |         |
| 189         | 193 | 21 | Male   | 2nd Class | yes | 75 | Pune    |
| Mediocre    |     |    |        |           |     |    |         |
| 190         | 194 | 19 | Male   | 2nd Class | no  | 74 | Chennai |
| Mediocre    |     |    |        |           |     |    |         |
| 191         | 195 | 20 | Female | 3rd Class | yes | 43 | Delhi   |
| Weak        |     |    |        |           |     |    |         |
| 192         | 196 | 19 | Male   | 2nd Class | no  | 74 | Mumbai  |
| Mediocre    |     |    |        |           |     |    |         |
| 193         | 197 | 21 | Female | 2nd Class | yes | 74 | Pune    |
| Mediocre    |     |    |        |           |     |    |         |
| 194         | 198 | 22 | Male   | 3rd Class | no  | 53 | Kochi   |
| Mediocre    |     |    |        |           |     |    |         |
| 195         | 199 | 21 | Female | 1st Class | no  | 87 | Gwalior |
| Intelligent |     |    |        |           |     |    |         |
| 196         | 200 | 20 | Male   | 2nd Class | no  | 75 | Bhopal  |
| Mediocre    |     |    |        |           |     |    |         |
| 197         | 201 | 20 | Male   | 3rd Class | no  | 42 | Chennai |
| Weak        |     |    |        |           |     |    |         |
| 198         | 202 | 19 | Female | 1st Class | no  | 80 | Delhi   |
| Mediocre    |     |    |        |           |     |    |         |

|                    |     |    |        |           |     |    |         |
|--------------------|-----|----|--------|-----------|-----|----|---------|
| 199<br>Weak        | 203 | 20 | Male   | 3rd Class | no  | 46 | Mumbai  |
| 200<br>Intelligent | 204 | 19 | Male   | 1st Class | yes | 91 | Pune    |
| 201<br>Mediocre    | 205 | 20 | Female | 2nd Class | no  | 78 | Kochi   |
| 202<br>Mediocre    | 206 | 18 | Male   | 1st Class | no  | 80 | Gwalior |
| 203<br>Mediocre    | 207 | 21 | Female | 2nd Class | no  | 79 | Chennai |
| 204<br>Intelligent | 208 | 19 | Male   | 1st Class | no  | 91 | Delhi   |
| 205<br>Mediocre    | 209 | 20 | Male   | 2nd Class | yes | 70 | Mumbai  |
| 206<br>Mediocre    | 210 | 19 | Female | 3rd Class | yes | 59 | Pune    |
| 207<br>Mediocre    | 211 | 21 | Male   | 3rd Class | yes | 58 | Kochi   |
| 208<br>Mediocre    | 212 | 22 | Female | 3rd Class | yes | 56 | Gwalior |
| 209<br>Intelligent | 213 | 21 | Male   | 1st Class | no  | 92 | Bhopal  |
| 210<br>Mediocre    | 214 | 20 | Female | 3rd Class | yes | 54 | Chennai |
| 211<br>Intelligent | 215 | 20 | Male   | 1st Class | no  | 93 | Delhi   |
| 212<br>Intelligent | 216 | 19 | Male   | 1st Class | yes | 95 | Mumbai  |
| 213<br>Mediocre    | 217 | 20 | Female | 3rd Class | no  | 53 | Pune    |
| 214<br>Intelligent | 218 | 19 | Male   | 1st Class | yes | 96 | Kochi   |
| 215<br>Mediocre    | 219 | 19 | Female | 2nd Class | no  | 75 | Gwalior |
| 216<br>Mediocre    | 220 | 20 | Male   | 2nd Class | yes | 75 | Bhopal  |
| 217<br>Mediocre    | 221 | 18 | Female | 3rd Class | no  | 52 | Pune    |
| 218<br>Mediocre    | 222 | 21 | Male   | 2nd Class | yes | 74 | Kochi   |
| 219<br>Mediocre    | 223 | 19 | Male   | 2nd Class | no  | 79 | Gwalior |
| 220<br>Mediocre    | 224 | 20 | Female | 3rd Class | yes | 50 | Bhopal  |
| 221<br>Mediocre    | 225 | 19 | Male   | 2nd Class | no  | 72 | Chennai |
| 222<br>Mediocre    | 226 | 21 | Female | 2nd Class | yes | 71 | Delhi   |
| 223<br>Mediocre    | 227 | 22 | Male   | 3rd Class | no  | 51 | Mumbai  |

|             |     |    |        |           |     |    |         |
|-------------|-----|----|--------|-----------|-----|----|---------|
| 224         | 228 | 21 | Female | 1st Class | no  | 99 | Pune    |
| Intelligent |     |    |        |           |     |    |         |
| 225         | 229 | 20 | Male   | 2nd Class | no  | 70 | Chennai |
| Mediocre    |     |    |        |           |     |    |         |
| 226         | 230 | 20 | Male   | 3rd Class | yes | 55 | Delhi   |
| Mediocre    |     |    |        |           |     |    |         |
| 227         | 231 | 19 | Female | 1st Class | yes | 97 | Mumbai  |
| Intelligent |     |    |        |           |     |    |         |
| 228         | 232 | 20 | Male   | 3rd Class | yes | 59 | Pune    |
| Mediocre    |     |    |        |           |     |    |         |

*##Q2.f) males and females count from each city*

```
gender_count_by_city = combined_df.groupby(['City',
'Gender']).size().unstack(fill_value=0)
gender_count_by_city
```

| Gender  | Female | Male |
|---------|--------|------|
| City    |        |      |
| Bhopal  | 13     | 15   |
| Chennai | 15     | 18   |
| Delhi   | 12     | 20   |
| Gwalior | 14     | 18   |
| Kochi   | 10     | 21   |
| Mumbai  | 16     | 16   |
| Pune    | 15     | 26   |

*##Q2.g) top 5 male scorers*

```
male_students = combined_df[combined_df['Gender'] == 'Male']
```

```
top_5_male_scorers = male_students.sort_values(by='Mark',
ascending=False).head(5)
```

```
print(top_5_male_scorers[['Student_id', 'Mark']])
```

|     | Student_id | Mark |
|-----|------------|------|
| 146 | 150        | 100  |
| 144 | 148        | 99   |
| 57  | 61         | 98   |
| 59  | 63         | 98   |
| 98  | 102        | 98   |

*##Q2.h) Replace gender with M & F and export to D drive*

```
combined_df['Gender'] = combined_df['Gender'].replace({'Male': 'M',
'Female': 'F'})
```

```
print(combined_df)
```

```
output_path = "D:/test.csv"
```

```
combined_df.to_csv(output_path, index=False)
```

|             | Student_id | Age | Gender | Grade     | Employed | Mark | City    |
|-------------|------------|-----|--------|-----------|----------|------|---------|
| IQ_level    |            |     |        |           |          |      |         |
| 0           | 1          | 19  | M      | 1st Class | yes      | 95   | Chennai |
| Intelligent |            |     |        |           |          |      |         |

|             |    |    |   |           |     |    |         |
|-------------|----|----|---|-----------|-----|----|---------|
| 1           | 2  | 20 | F | 2nd Class | no  | 70 | Delhi   |
| Mediocre    |    |    |   |           |     |    |         |
| 2           | 3  | 18 | M | 1st Class | no  | 98 | Mumbai  |
| Intelligent |    |    |   |           |     |    |         |
| 3           | 4  | 21 | F | 2nd Class | no  | 75 | Pune    |
| Mediocre    |    |    |   |           |     |    |         |
| 4           | 5  | 19 | M | 1st Class | no  | 89 | Kochi   |
| Intelligent |    |    |   |           |     |    |         |
| 5           | 6  | 20 | M | 2nd Class | yes | 69 | Gwalior |
| Mediocre    |    |    |   |           |     |    |         |
| 6           | 7  | 19 | F | 3rd Class | yes | 52 | Bhopal  |
| Mediocre    |    |    |   |           |     |    |         |
| 7           | 8  | 21 | M | 3rd Class | yes | 54 | Chennai |
| Mediocre    |    |    |   |           |     |    |         |
| 8           | 11 | 20 | F | 3rd Class | yes | 52 | Pune    |
| Mediocre    |    |    |   |           |     |    |         |
| 9           | 12 | 20 | M | 1st Class | no  | 90 | Kochi   |
| Intelligent |    |    |   |           |     |    |         |
| 10          | 13 | 19 | M | 1st Class | yes | 92 | Gwalior |
| Intelligent |    |    |   |           |     |    |         |
| 11          | 14 | 20 | F | 3rd Class | no  | 55 | Bhopal  |
| Mediocre    |    |    |   |           |     |    |         |
| 12          | 15 | 19 | M | 1st Class | yes | 97 | Pune    |
| Intelligent |    |    |   |           |     |    |         |
| 13          | 16 | 19 | F | 2nd Class | no  | 77 | Kochi   |
| Mediocre    |    |    |   |           |     |    |         |
| 14          | 17 | 20 | M | 2nd Class | yes | 68 | Gwalior |
| Mediocre    |    |    |   |           |     |    |         |
| 15          | 18 | 18 | F | 3rd Class | no  | 51 | Bhopal  |
| Mediocre    |    |    |   |           |     |    |         |
| 16          | 19 | 21 | M | 2nd Class | yes | 72 | Chennai |
| Mediocre    |    |    |   |           |     |    |         |
| 17          | 20 | 19 | M | 2nd Class | no  | 79 | Delhi   |
| Mediocre    |    |    |   |           |     |    |         |
| 18          | 21 | 20 | F | 3rd Class | yes | 49 | Mumbai  |
| Weak        |    |    |   |           |     |    |         |
| 19          | 22 | 19 | M | 2nd Class | no  | 78 | Pune    |
| Mediocre    |    |    |   |           |     |    |         |
| 20          | 23 | 21 | F | 2nd Class | yes | 76 | Chennai |
| Mediocre    |    |    |   |           |     |    |         |
| 21          | 24 | 22 | M | 3rd Class | no  | 43 | Delhi   |
| Weak        |    |    |   |           |     |    |         |
| 22          | 25 | 21 | F | 1st Class | no  | 89 | Mumbai  |
| Intelligent |    |    |   |           |     |    |         |
| 23          | 26 | 20 | M | 2nd Class | no  | 75 | Pune    |
| Mediocre    |    |    |   |           |     |    |         |
| 24          | 27 | 20 | M | 3rd Class | no  | 45 | Kochi   |
| Weak        |    |    |   |           |     |    |         |
| 25          | 28 | 19 | F | 1st Class | no  | 98 | Gwalior |

|             |    |    |    |   |           |     |    |         |
|-------------|----|----|----|---|-----------|-----|----|---------|
| Intelligent | 26 | 29 | 20 | M | 3rd Class | no  | 54 | Bhopal  |
| Mediocre    | 27 | 30 | 19 | M | 1st Class | yes | 81 | Chennai |
| Intelligent | 28 | 31 | 20 | F | 2nd Class | no  | 74 | Delhi   |
| Mediocre    | 29 | 32 | 18 | M | 1st Class | no  | 82 | Mumbai  |
| Intelligent | 30 | 33 | 21 | F | 2nd Class | no  | 78 | Pune    |
| Mediocre    | 31 | 34 | 19 | M | 1st Class | no  | 83 | Kochi   |
| Intelligent | 32 | 35 | 20 | M | 2nd Class | yes | 79 | Gwalior |
| Mediocre    | 33 | 36 | 19 | F | 3rd Class | yes | 55 | Bhopal  |
| Mediocre    | 34 | 37 | 21 | M | 3rd Class | yes | 53 | Pune    |
| Mediocre    | 35 | 38 | 22 | F | 3rd Class | yes | 55 | Kochi   |
| Mediocre    | 36 | 39 | 21 | M | 1st Class | no  | 84 | Gwalior |
| Intelligent | 37 | 40 | 20 | F | 3rd Class | yes | 44 | Bhopal  |
| Weak        | 38 | 41 | 20 | M | 1st Class | no  | 85 | Chennai |
| Intelligent | 39 | 42 | 19 | M | 1st Class | yes | 86 | Delhi   |
| Intelligent | 40 | 43 | 20 | F | 3rd Class | no  | 45 | Mumbai  |
| Weak        | 41 | 44 | 19 | M | 1st Class | yes | 95 | Pune    |
| Intelligent | 42 | 45 | 19 | F | 2nd Class | no  | 73 | Chennai |
| Mediocre    | 43 | 46 | 20 | M | 2nd Class | yes | 79 | Delhi   |
| Mediocre    | 44 | 47 | 18 | F | 3rd Class | no  | 46 | Mumbai  |
| Weak        | 45 | 48 | 21 | M | 2nd Class | yes | 67 | Pune    |
| Mediocre    | 46 | 50 | 20 | F | 3rd Class | yes | 47 | Gwalior |
| Weak        | 47 | 51 | 19 | M | 2nd Class | no  | 65 | Bhopal  |
| Mediocre    | 48 | 52 | 21 | F | 2nd Class | yes | 67 | Chennai |
| Mediocre    | 49 | 53 | 22 | M | 3rd Class | no  | 49 | Delhi   |
| Weak        |    |    |    |   |           |     |    |         |

|             |    |    |   |           |     |    |         |
|-------------|----|----|---|-----------|-----|----|---------|
| 50          | 54 | 21 | F | 1st Class | no  | 82 | Mumbai  |
| Intelligent |    |    |   |           |     |    |         |
| 51          | 55 | 20 | M | 2nd Class | no  | 69 | Pune    |
| Mediocre    |    |    |   |           |     |    |         |
| 52          | 56 | 20 | M | 3rd Class | no  | 48 | Kochi   |
| Weak        |    |    |   |           |     |    |         |
| 53          | 57 | 19 | F | 1st Class | no  | 85 | Gwalior |
| Intelligent |    |    |   |           |     |    |         |
| 54          | 58 | 20 | M | 3rd Class | no  | 42 | Bhopal  |
| Weak        |    |    |   |           |     |    |         |
| 55          | 59 | 19 | M | 1st Class | yes | 89 | Pune    |
| Intelligent |    |    |   |           |     |    |         |
| 56          | 60 | 20 | F | 2nd Class | no  | 66 | Kochi   |
| Mediocre    |    |    |   |           |     |    |         |
| 57          | 61 | 18 | M | 1st Class | no  | 98 | Gwalior |
| Intelligent |    |    |   |           |     |    |         |
| 58          | 62 | 21 | F | 2nd Class | no  | 77 | Bhopal  |
| Mediocre    |    |    |   |           |     |    |         |
| 59          | 63 | 19 | M | 1st Class | no  | 98 | Chennai |
| Intelligent |    |    |   |           |     |    |         |
| 60          | 64 | 20 | M | 2nd Class | yes | 78 | Delhi   |
| Mediocre    |    |    |   |           |     |    |         |
| 61          | 65 | 19 | F | 3rd Class | yes | 43 | Mumbai  |
| Weak        |    |    |   |           |     |    |         |
| 62          | 66 | 21 | M | 3rd Class | yes | 59 | Pune    |
| Mediocre    |    |    |   |           |     |    |         |
| 63          | 67 | 22 | F | 3rd Class | yes | 58 | Chennai |
| Mediocre    |    |    |   |           |     |    |         |
| 64          | 68 | 21 | M | 1st Class | no  | 87 | Delhi   |
| Intelligent |    |    |   |           |     |    |         |
| 65          | 69 | 20 | F | 3rd Class | yes | 55 | Mumbai  |
| Mediocre    |    |    |   |           |     |    |         |
| 66          | 70 | 20 | M | 1st Class | no  | 86 | Pune    |
| Intelligent |    |    |   |           |     |    |         |
| 67          | 71 | 19 | M | 1st Class | yes | 85 | Kochi   |
| Intelligent |    |    |   |           |     |    |         |
| 68          | 72 | 20 | F | 3rd Class | no  | 54 | Gwalior |
| Mediocre    |    |    |   |           |     |    |         |
| 69          | 73 | 19 | M | 1st Class | yes | 89 | Bhopal  |
| Intelligent |    |    |   |           |     |    |         |
| 70          | 74 | 19 | F | 2nd Class | no  | 77 | Chennai |
| Mediocre    |    |    |   |           |     |    |         |
| 71          | 75 | 20 | M | 2nd Class | yes | 79 | Delhi   |
| Mediocre    |    |    |   |           |     |    |         |
| 72          | 76 | 18 | F | 3rd Class | no  | 53 | Mumbai  |
| Mediocre    |    |    |   |           |     |    |         |
| 73          | 77 | 21 | M | 2nd Class | yes | 77 | Pune    |
| Mediocre    |    |    |   |           |     |    |         |
| 74          | 78 | 19 | M | 2nd Class | no  | 76 | Kochi   |



|             |     |    |   |           |     |    |         |  |
|-------------|-----|----|---|-----------|-----|----|---------|--|
| Mediocre    |     |    |   |           |     |    |         |  |
| 75          | 79  | 20 | F | 3rd Class | yes | 56 | Gwalior |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 76          | 80  | 19 | M | 2nd Class | no  | 75 | Bhopal  |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 77          | 81  | 21 | F | 2nd Class | yes | 74 | Pune    |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 78          | 82  | 22 | M | 3rd Class | no  | 59 | Kochi   |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 79          | 83  | 21 | F | 1st Class | no  | 98 | Gwalior |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 80          | 84  | 20 | M | 2nd Class | no  | 73 | Bhopal  |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 81          | 85  | 20 | M | 3rd Class | no  | 58 | Chennai |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 82          | 86  | 19 | F | 1st Class | no  | 97 | Delhi   |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 83          | 87  | 20 | M | 3rd Class | no  | 57 | Mumbai  |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 84          | 88  | 19 | M | 1st Class | yes | 96 | Pune    |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 85          | 89  | 20 | F | 2nd Class | no  | 75 | Chennai |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 86          | 90  | 18 | M | 1st Class | no  | 94 | Delhi   |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 87          | 91  | 21 | F | 2nd Class | no  | 74 | Mumbai  |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 88          | 92  | 19 | M | 1st Class | no  | 93 | Pune    |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 89          | 93  | 20 | M | 2nd Class | yes | 74 | Kochi   |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 90          | 94  | 19 | F | 3rd Class | yes | 53 | Gwalior |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 91          | 95  | 21 | M | 3rd Class | yes | 50 | Bhopal  |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 92          | 96  | 22 | F | 3rd Class | yes | 40 | Chennai |  |
| Weak        |     |    |   |           |     |    |         |  |
| 93          | 97  | 21 | M | 1st Class | no  | 87 | Delhi   |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 94          | 98  | 20 | F | 3rd Class | yes | 44 | Mumbai  |  |
| Weak        |     |    |   |           |     |    |         |  |
| 95          | 99  | 20 | M | 1st Class | no  | 86 | Pune    |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 96          | 100 | 19 | M | 1st Class | yes | 84 | Kochi   |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 97          | 101 | 20 | F | 3rd Class | no  | 43 | Gwalior |  |
| Weak        |     |    |   |           |     |    |         |  |
| 98          | 102 | 19 | M | 1st Class | yes | 98 | Chennai |  |
| Intelligent |     |    |   |           |     |    |         |  |

|             |     |    |   |           |     |    |         |
|-------------|-----|----|---|-----------|-----|----|---------|
| 99          | 103 | 19 | F | 2nd Class | no  | 75 | Delhi   |
| Mediocre    |     |    |   |           |     |    |         |
| 100         | 104 | 20 | M | 2nd Class | yes | 78 | Mumbai  |
| Mediocre    |     |    |   |           |     |    |         |
| 101         | 105 | 18 | F | 3rd Class | no  | 42 | Pune    |
| Weak        |     |    |   |           |     |    |         |
| 102         | 106 | 21 | M | 2nd Class | yes | 77 | Kochi   |
| Mediocre    |     |    |   |           |     |    |         |
| 103         | 107 | 19 | M | 2nd Class | no  | 78 | Gwalior |
| Mediocre    |     |    |   |           |     |    |         |
| 104         | 108 | 20 | F | 3rd Class | yes | 41 | Bhopal  |
| Weak        |     |    |   |           |     |    |         |
| 105         | 109 | 19 | M | 2nd Class | no  | 79 | Chennai |
| Mediocre    |     |    |   |           |     |    |         |
| 106         | 110 | 21 | F | 2nd Class | yes | 76 | Delhi   |
| Mediocre    |     |    |   |           |     |    |         |
| 107         | 111 | 22 | M | 3rd Class | no  | 45 | Mumbai  |
| Weak        |     |    |   |           |     |    |         |
| 108         | 112 | 21 | F | 1st Class | no  | 97 | Pune    |
| Intelligent |     |    |   |           |     |    |         |
| 109         | 113 | 20 | M | 2nd Class | no  | 75 | Kochi   |
| Mediocre    |     |    |   |           |     |    |         |
| 110         | 114 | 20 | M | 3rd Class | no  | 46 | Gwalior |
| Weak        |     |    |   |           |     |    |         |
| 111         | 115 | 19 | F | 1st Class | no  | 96 | Bhopal  |
| Intelligent |     |    |   |           |     |    |         |
| 112         | 116 | 20 | M | 3rd Class | no  | 47 | Pune    |
| Weak        |     |    |   |           |     |    |         |
| 113         | 117 | 19 | M | 1st Class | yes | 90 | Kochi   |
| Intelligent |     |    |   |           |     |    |         |
| 114         | 118 | 20 | F | 2nd Class | no  | 79 | Gwalior |
| Mediocre    |     |    |   |           |     |    |         |
| 115         | 119 | 18 | M | 1st Class | no  | 91 | Bhopal  |
| Intelligent |     |    |   |           |     |    |         |
| 116         | 120 | 21 | F | 2nd Class | no  | 77 | Chennai |
| Mediocre    |     |    |   |           |     |    |         |
| 117         | 121 | 19 | M | 1st Class | no  | 87 | Delhi   |
| Intelligent |     |    |   |           |     |    |         |
| 118         | 122 | 20 | M | 2nd Class | yes | 75 | Mumbai  |
| Mediocre    |     |    |   |           |     |    |         |
| 119         | 123 | 19 | F | 3rd Class | yes | 48 | Pune    |
| Weak        |     |    |   |           |     |    |         |
| 120         | 124 | 21 | M | 3rd Class | yes | 49 | Chennai |
| Weak        |     |    |   |           |     |    |         |
| 121         | 125 | 22 | F | 3rd Class | yes | 55 | Delhi   |
| Mediocre    |     |    |   |           |     |    |         |
| 122         | 126 | 21 | M | 1st Class | no  | 88 | Mumbai  |
| Intelligent |     |    |   |           |     |    |         |
| 123         | 127 | 20 | F | 3rd Class | yes | 56 | Pune    |

|             |     |    |   |           |     |     |         |  |
|-------------|-----|----|---|-----------|-----|-----|---------|--|
| Mediocre    |     |    |   |           |     |     |         |  |
| 124         | 128 | 20 | M | 1st Class | no  | 86  | Kochi   |  |
| Intelligent |     |    |   |           |     |     |         |  |
| 125         | 129 | 19 | M | 1st Class | yes | 89  | Gwalior |  |
| Intelligent |     |    |   |           |     |     |         |  |
| 126         | 130 | 20 | F | 3rd Class | no  | 57  | Bhopal  |  |
| Mediocre    |     |    |   |           |     |     |         |  |
| 127         | 131 | 19 | M | 1st Class | yes | 90  | Chennai |  |
| Intelligent |     |    |   |           |     |     |         |  |
| 128         | 132 | 19 | F | 2nd Class | no  | 75  | Delhi   |  |
| Mediocre    |     |    |   |           |     |     |         |  |
| 129         | 133 | 20 | M | 2nd Class | yes | 75  | Mumbai  |  |
| Mediocre    |     |    |   |           |     |     |         |  |
| 130         | 134 | 18 | F | 3rd Class | no  | 58  | Pune    |  |
| Mediocre    |     |    |   |           |     |     |         |  |
| 131         | 135 | 21 | M | 2nd Class | yes | 75  | Kochi   |  |
| Mediocre    |     |    |   |           |     |     |         |  |
| 132         | 136 | 19 | M | 2nd Class | no  | 74  | Gwalior |  |
| Mediocre    |     |    |   |           |     |     |         |  |
| 133         | 137 | 20 | F | 3rd Class | yes | 59  | Chennai |  |
| Mediocre    |     |    |   |           |     |     |         |  |
| 134         | 138 | 19 | M | 2nd Class | no  | 73  | Delhi   |  |
| Mediocre    |     |    |   |           |     |     |         |  |
| 135         | 139 | 21 | F | 2nd Class | yes | 72  | Mumbai  |  |
| Mediocre    |     |    |   |           |     |     |         |  |
| 136         | 140 | 22 | M | 3rd Class | no  | 50  | Pune    |  |
| Mediocre    |     |    |   |           |     |     |         |  |
| 137         | 141 | 21 | F | 1st Class | no  | 93  | Kochi   |  |
| Intelligent |     |    |   |           |     |     |         |  |
| 138         | 142 | 20 | M | 2nd Class | no  | 71  | Gwalior |  |
| Mediocre    |     |    |   |           |     |     |         |  |
| 139         | 143 | 20 | M | 3rd Class | no  | 40  | Bhopal  |  |
| Weak        |     |    |   |           |     |     |         |  |
| 140         | 144 | 19 | F | 1st Class | no  | 97  | Chennai |  |
| Intelligent |     |    |   |           |     |     |         |  |
| 141         | 145 | 20 | M | 3rd Class | no  | 43  | Delhi   |  |
| Weak        |     |    |   |           |     |     |         |  |
| 142         | 146 | 19 | M | 1st Class | yes | 98  | Mumbai  |  |
| Intelligent |     |    |   |           |     |     |         |  |
| 143         | 147 | 20 | F | 2nd Class | no  | 70  | Pune    |  |
| Mediocre    |     |    |   |           |     |     |         |  |
| 144         | 148 | 18 | M | 1st Class | no  | 99  | Kochi   |  |
| Intelligent |     |    |   |           |     |     |         |  |
| 145         | 149 | 21 | F | 2nd Class | no  | 77  | Gwalior |  |
| Mediocre    |     |    |   |           |     |     |         |  |
| 146         | 150 | 19 | M | 1st Class | no  | 100 | Bhopal  |  |
| Intelligent |     |    |   |           |     |     |         |  |
| 147         | 151 | 20 | M | 2nd Class | yes | 78  | Pune    |  |
| Mediocre    |     |    |   |           |     |     |         |  |

|             |     |    |   |           |     |    |         |
|-------------|-----|----|---|-----------|-----|----|---------|
| 148         | 152 | 19 | F | 3rd Class | yes | 45 | Kochi   |
| Weak        |     |    |   |           |     |    |         |
| 149         | 153 | 21 | M | 3rd Class | yes | 46 | Gwalior |
| Weak        |     |    |   |           |     |    |         |
| 150         | 154 | 22 | F | 3rd Class | yes | 47 | Bhopal  |
| Weak        |     |    |   |           |     |    |         |
| 151         | 155 | 21 | M | 1st Class | no  | 89 | Chennai |
| Intelligent |     |    |   |           |     |    |         |
| 152         | 156 | 20 | F | 3rd Class | yes | 49 | Delhi   |
| Weak        |     |    |   |           |     |    |         |
| 153         | 157 | 20 | M | 1st Class | no  | 86 | Mumbai  |
| Intelligent |     |    |   |           |     |    |         |
| 154         | 158 | 19 | M | 1st Class | yes | 92 | Pune    |
| Intelligent |     |    |   |           |     |    |         |
| 155         | 159 | 20 | F | 3rd Class | no  | 50 | Chennai |
| Mediocre    |     |    |   |           |     |    |         |
| 156         | 160 | 19 | M | 1st Class | yes | 92 | Delhi   |
| Intelligent |     |    |   |           |     |    |         |
| 157         | 161 | 19 | F | 2nd Class | no  | 79 | Mumbai  |
| Mediocre    |     |    |   |           |     |    |         |
| 158         | 162 | 20 | M | 2nd Class | yes | 79 | Pune    |
| Mediocre    |     |    |   |           |     |    |         |
| 159         | 163 | 18 | F | 3rd Class | no  | 59 | Kochi   |
| Mediocre    |     |    |   |           |     |    |         |
| 160         | 164 | 21 | M | 2nd Class | yes | 75 | Gwalior |
| Mediocre    |     |    |   |           |     |    |         |
| 161         | 165 | 19 | M | 2nd Class | no  | 75 | Bhopal  |
| Mediocre    |     |    |   |           |     |    |         |
| 162         | 166 | 20 | F | 3rd Class | yes | 53 | Chennai |
| Mediocre    |     |    |   |           |     |    |         |
| 163         | 167 | 19 | M | 2nd Class | no  | 74 | Delhi   |
| Mediocre    |     |    |   |           |     |    |         |
| 164         | 168 | 21 | F | 2nd Class | yes | 74 | Mumbai  |
| Mediocre    |     |    |   |           |     |    |         |
| 165         | 169 | 22 | M | 3rd Class | no  | 53 | Pune    |
| Mediocre    |     |    |   |           |     |    |         |
| 166         | 170 | 21 | F | 1st Class | no  | 90 | Kochi   |
| Intelligent |     |    |   |           |     |    |         |
| 167         | 171 | 20 | M | 2nd Class | no  | 73 | Gwalior |
| Mediocre    |     |    |   |           |     |    |         |
| 168         | 172 | 20 | M | 3rd Class | no  | 52 | Chennai |
| Mediocre    |     |    |   |           |     |    |         |
| 169         | 173 | 19 | F | 1st Class | no  | 98 | Delhi   |
| Intelligent |     |    |   |           |     |    |         |
| 170         | 174 | 20 | M | 3rd Class | no  | 56 | Mumbai  |
| Mediocre    |     |    |   |           |     |    |         |
| 171         | 175 | 19 | M | 1st Class | yes | 84 | Pune    |
| Intelligent |     |    |   |           |     |    |         |
| 172         | 176 | 20 | F | 2nd Class | no  | 77 | Kochi   |

|             |     |    |   |           |     |    |         |  |
|-------------|-----|----|---|-----------|-----|----|---------|--|
| Mediocre    |     |    |   |           |     |    |         |  |
| 173         | 177 | 18 | M | 1st Class | no  | 97 | Gwalior |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 174         | 178 | 21 | F | 2nd Class | no  | 76 | Bhopal  |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 175         | 179 | 19 | M | 1st Class | no  | 94 | Chennai |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 176         | 180 | 20 | M | 2nd Class | yes | 76 | Delhi   |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 177         | 181 | 19 | F | 3rd Class | yes | 56 | Mumbai  |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 178         | 182 | 21 | M | 3rd Class | yes | 58 | Pune    |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 179         | 183 | 22 | F | 3rd Class | yes | 59 | Kochi   |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 180         | 184 | 21 | M | 1st Class | no  | 87 | Gwalior |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 181         | 185 | 20 | F | 3rd Class | yes | 50 | Bhopal  |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 182         | 186 | 20 | M | 1st Class | no  | 85 | Pune    |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 183         | 187 | 19 | M | 1st Class | yes | 83 | Kochi   |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 184         | 188 | 20 | F | 3rd Class | no  | 40 | Gwalior |  |
| Weak        |     |    |   |           |     |    |         |  |
| 185         | 189 | 19 | M | 1st Class | yes | 89 | Bhopal  |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 186         | 190 | 19 | F | 2nd Class | no  | 75 | Chennai |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 187         | 191 | 20 | M | 2nd Class | yes | 77 | Delhi   |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 188         | 192 | 18 | F | 3rd Class | no  | 44 | Mumbai  |  |
| Weak        |     |    |   |           |     |    |         |  |
| 189         | 193 | 21 | M | 2nd Class | yes | 75 | Pune    |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 190         | 194 | 19 | M | 2nd Class | no  | 74 | Chennai |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 191         | 195 | 20 | F | 3rd Class | yes | 43 | Delhi   |  |
| Weak        |     |    |   |           |     |    |         |  |
| 192         | 196 | 19 | M | 2nd Class | no  | 74 | Mumbai  |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 193         | 197 | 21 | F | 2nd Class | yes | 74 | Pune    |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 194         | 198 | 22 | M | 3rd Class | no  | 53 | Kochi   |  |
| Mediocre    |     |    |   |           |     |    |         |  |
| 195         | 199 | 21 | F | 1st Class | no  | 87 | Gwalior |  |
| Intelligent |     |    |   |           |     |    |         |  |
| 196         | 200 | 20 | M | 2nd Class | no  | 75 | Bhopal  |  |
| Mediocre    |     |    |   |           |     |    |         |  |

|             |     |    |   |           |     |    |         |
|-------------|-----|----|---|-----------|-----|----|---------|
| 197         | 201 | 20 | M | 3rd Class | no  | 42 | Chennai |
| Weak        |     |    |   |           |     |    |         |
| 198         | 202 | 19 | F | 1st Class | no  | 80 | Delhi   |
| Mediocre    |     |    |   |           |     |    |         |
| 199         | 203 | 20 | M | 3rd Class | no  | 46 | Mumbai  |
| Weak        |     |    |   |           |     |    |         |
| 200         | 204 | 19 | M | 1st Class | yes | 91 | Pune    |
| Intelligent |     |    |   |           |     |    |         |
| 201         | 205 | 20 | F | 2nd Class | no  | 78 | Kochi   |
| Mediocre    |     |    |   |           |     |    |         |
| 202         | 206 | 18 | M | 1st Class | no  | 80 | Gwalior |
| Mediocre    |     |    |   |           |     |    |         |
| 203         | 207 | 21 | F | 2nd Class | no  | 79 | Chennai |
| Mediocre    |     |    |   |           |     |    |         |
| 204         | 208 | 19 | M | 1st Class | no  | 91 | Delhi   |
| Intelligent |     |    |   |           |     |    |         |
| 205         | 209 | 20 | M | 2nd Class | yes | 70 | Mumbai  |
| Mediocre    |     |    |   |           |     |    |         |
| 206         | 210 | 19 | F | 3rd Class | yes | 59 | Pune    |
| Mediocre    |     |    |   |           |     |    |         |
| 207         | 211 | 21 | M | 3rd Class | yes | 58 | Kochi   |
| Mediocre    |     |    |   |           |     |    |         |
| 208         | 212 | 22 | F | 3rd Class | yes | 56 | Gwalior |
| Mediocre    |     |    |   |           |     |    |         |
| 209         | 213 | 21 | M | 1st Class | no  | 92 | Bhopal  |
| Intelligent |     |    |   |           |     |    |         |
| 210         | 214 | 20 | F | 3rd Class | yes | 54 | Chennai |
| Mediocre    |     |    |   |           |     |    |         |
| 211         | 215 | 20 | M | 1st Class | no  | 93 | Delhi   |
| Intelligent |     |    |   |           |     |    |         |
| 212         | 216 | 19 | M | 1st Class | yes | 95 | Mumbai  |
| Intelligent |     |    |   |           |     |    |         |
| 213         | 217 | 20 | F | 3rd Class | no  | 53 | Pune    |
| Mediocre    |     |    |   |           |     |    |         |
| 214         | 218 | 19 | M | 1st Class | yes | 96 | Kochi   |
| Intelligent |     |    |   |           |     |    |         |
| 215         | 219 | 19 | F | 2nd Class | no  | 75 | Gwalior |
| Mediocre    |     |    |   |           |     |    |         |
| 216         | 220 | 20 | M | 2nd Class | yes | 75 | Bhopal  |
| Mediocre    |     |    |   |           |     |    |         |
| 217         | 221 | 18 | F | 3rd Class | no  | 52 | Pune    |
| Mediocre    |     |    |   |           |     |    |         |
| 218         | 222 | 21 | M | 2nd Class | yes | 74 | Kochi   |
| Mediocre    |     |    |   |           |     |    |         |
| 219         | 223 | 19 | M | 2nd Class | no  | 79 | Gwalior |
| Mediocre    |     |    |   |           |     |    |         |
| 220         | 224 | 20 | F | 3rd Class | yes | 50 | Bhopal  |
| Mediocre    |     |    |   |           |     |    |         |
| 221         | 225 | 19 | M | 2nd Class | no  | 72 | Chennai |

|             |     |     |    |   |           |     |    |         |
|-------------|-----|-----|----|---|-----------|-----|----|---------|
| Mediocre    | 222 | 226 | 21 | F | 2nd Class | yes | 71 | Delhi   |
| Mediocre    | 223 | 227 | 22 | M | 3rd Class | no  | 51 | Mumbai  |
| Mediocre    | 224 | 228 | 21 | F | 1st Class | no  | 99 | Pune    |
| Intelligent | 225 | 229 | 20 | M | 2nd Class | no  | 70 | Chennai |
| Mediocre    | 226 | 230 | 20 | M | 3rd Class | yes | 55 | Delhi   |
| Mediocre    | 227 | 231 | 19 | F | 1st Class | yes | 97 | Mumbai  |
| Intelligent | 228 | 232 | 20 | M | 3rd Class | yes | 59 | Pune    |
| Mediocre    |     |     |    |   |           |     |    |         |

*##Q2.i) check for Student\_id duplications*

```
duplicate_ids =
combined_df[combined_df.duplicated(subset=['Student_id'], keep=False)]

print(duplicate_ids)
```

Empty DataFrame

Columns: [Student\_id, Age, Gender, Grade, Employed, Mark, City, IQ\_level]  
Index: []

*##Q2.j) separate dataframe for Integer/Float variables*

```
numeric_df = combined_df.select_dtypes(include=['int64', 'float64'])
```

numeric\_df

|    | Student_id | Age | Mark |
|----|------------|-----|------|
| 0  | 1          | 19  | 95   |
| 1  | 2          | 20  | 70   |
| 2  | 3          | 18  | 98   |
| 3  | 4          | 21  | 75   |
| 4  | 5          | 19  | 89   |
| 5  | 6          | 20  | 69   |
| 6  | 7          | 19  | 52   |
| 7  | 8          | 21  | 54   |
| 8  | 11         | 20  | 52   |
| 9  | 12         | 20  | 90   |
| 10 | 13         | 19  | 92   |
| 11 | 14         | 20  | 55   |
| 12 | 15         | 19  | 97   |
| 13 | 16         | 19  | 77   |
| 14 | 17         | 20  | 68   |
| 15 | 18         | 18  | 51   |
| 16 | 19         | 21  | 72   |

|    |    |    |    |
|----|----|----|----|
| 17 | 20 | 19 | 79 |
| 18 | 21 | 20 | 49 |
| 19 | 22 | 19 | 78 |
| 20 | 23 | 21 | 76 |
| 21 | 24 | 22 | 43 |
| 22 | 25 | 21 | 89 |
| 23 | 26 | 20 | 75 |
| 24 | 27 | 20 | 45 |
| 25 | 28 | 19 | 98 |
| 26 | 29 | 20 | 54 |
| 27 | 30 | 19 | 81 |
| 28 | 31 | 20 | 74 |
| 29 | 32 | 18 | 82 |
| 30 | 33 | 21 | 78 |
| 31 | 34 | 19 | 83 |
| 32 | 35 | 20 | 79 |
| 33 | 36 | 19 | 55 |
| 34 | 37 | 21 | 53 |
| 35 | 38 | 22 | 55 |
| 36 | 39 | 21 | 84 |
| 37 | 40 | 20 | 44 |
| 38 | 41 | 20 | 85 |
| 39 | 42 | 19 | 86 |
| 40 | 43 | 20 | 45 |
| 41 | 44 | 19 | 95 |
| 42 | 45 | 19 | 73 |
| 43 | 46 | 20 | 79 |
| 44 | 47 | 18 | 46 |
| 45 | 48 | 21 | 67 |
| 46 | 50 | 20 | 47 |
| 47 | 51 | 19 | 65 |
| 48 | 52 | 21 | 67 |
| 49 | 53 | 22 | 49 |
| 50 | 54 | 21 | 82 |
| 51 | 55 | 20 | 69 |
| 52 | 56 | 20 | 48 |
| 53 | 57 | 19 | 85 |
| 54 | 58 | 20 | 42 |
| 55 | 59 | 19 | 89 |
| 56 | 60 | 20 | 66 |
| 57 | 61 | 18 | 98 |
| 58 | 62 | 21 | 77 |
| 59 | 63 | 19 | 98 |
| 60 | 64 | 20 | 78 |
| 61 | 65 | 19 | 43 |
| 62 | 66 | 21 | 59 |
| 63 | 67 | 22 | 58 |
| 64 | 68 | 21 | 87 |
| 65 | 69 | 20 | 55 |



|     |     |    |    |
|-----|-----|----|----|
| 66  | 70  | 20 | 86 |
| 67  | 71  | 19 | 85 |
| 68  | 72  | 20 | 54 |
| 69  | 73  | 19 | 89 |
| 70  | 74  | 19 | 77 |
| 71  | 75  | 20 | 79 |
| 72  | 76  | 18 | 53 |
| 73  | 77  | 21 | 77 |
| 74  | 78  | 19 | 76 |
| 75  | 79  | 20 | 56 |
| 76  | 80  | 19 | 75 |
| 77  | 81  | 21 | 74 |
| 78  | 82  | 22 | 59 |
| 79  | 83  | 21 | 98 |
| 80  | 84  | 20 | 73 |
| 81  | 85  | 20 | 58 |
| 82  | 86  | 19 | 97 |
| 83  | 87  | 20 | 57 |
| 84  | 88  | 19 | 96 |
| 85  | 89  | 20 | 75 |
| 86  | 90  | 18 | 94 |
| 87  | 91  | 21 | 74 |
| 88  | 92  | 19 | 93 |
| 89  | 93  | 20 | 74 |
| 90  | 94  | 19 | 53 |
| 91  | 95  | 21 | 50 |
| 92  | 96  | 22 | 40 |
| 93  | 97  | 21 | 87 |
| 94  | 98  | 20 | 44 |
| 95  | 99  | 20 | 86 |
| 96  | 100 | 19 | 84 |
| 97  | 101 | 20 | 43 |
| 98  | 102 | 19 | 98 |
| 99  | 103 | 19 | 75 |
| 100 | 104 | 20 | 78 |
| 101 | 105 | 18 | 42 |
| 102 | 106 | 21 | 77 |
| 103 | 107 | 19 | 78 |
| 104 | 108 | 20 | 41 |
| 105 | 109 | 19 | 79 |
| 106 | 110 | 21 | 76 |
| 107 | 111 | 22 | 45 |
| 108 | 112 | 21 | 97 |
| 109 | 113 | 20 | 75 |
| 110 | 114 | 20 | 46 |
| 111 | 115 | 19 | 96 |
| 112 | 116 | 20 | 47 |
| 113 | 117 | 19 | 90 |
| 114 | 118 | 20 | 79 |

|     |     |    |     |
|-----|-----|----|-----|
| 115 | 119 | 18 | 91  |
| 116 | 120 | 21 | 77  |
| 117 | 121 | 19 | 87  |
| 118 | 122 | 20 | 75  |
| 119 | 123 | 19 | 48  |
| 120 | 124 | 21 | 49  |
| 121 | 125 | 22 | 55  |
| 122 | 126 | 21 | 88  |
| 123 | 127 | 20 | 56  |
| 124 | 128 | 20 | 86  |
| 125 | 129 | 19 | 89  |
| 126 | 130 | 20 | 57  |
| 127 | 131 | 19 | 90  |
| 128 | 132 | 19 | 75  |
| 129 | 133 | 20 | 75  |
| 130 | 134 | 18 | 58  |
| 131 | 135 | 21 | 75  |
| 132 | 136 | 19 | 74  |
| 133 | 137 | 20 | 59  |
| 134 | 138 | 19 | 73  |
| 135 | 139 | 21 | 72  |
| 136 | 140 | 22 | 50  |
| 137 | 141 | 21 | 93  |
| 138 | 142 | 20 | 71  |
| 139 | 143 | 20 | 40  |
| 140 | 144 | 19 | 97  |
| 141 | 145 | 20 | 43  |
| 142 | 146 | 19 | 98  |
| 143 | 147 | 20 | 70  |
| 144 | 148 | 18 | 99  |
| 145 | 149 | 21 | 77  |
| 146 | 150 | 19 | 100 |
| 147 | 151 | 20 | 78  |
| 148 | 152 | 19 | 45  |
| 149 | 153 | 21 | 46  |
| 150 | 154 | 22 | 47  |
| 151 | 155 | 21 | 89  |
| 152 | 156 | 20 | 49  |
| 153 | 157 | 20 | 86  |
| 154 | 158 | 19 | 92  |
| 155 | 159 | 20 | 50  |
| 156 | 160 | 19 | 92  |
| 157 | 161 | 19 | 79  |
| 158 | 162 | 20 | 79  |
| 159 | 163 | 18 | 59  |
| 160 | 164 | 21 | 75  |
| 161 | 165 | 19 | 75  |
| 162 | 166 | 20 | 53  |
| 163 | 167 | 19 | 74  |

|     |     |    |    |
|-----|-----|----|----|
| 164 | 168 | 21 | 74 |
| 165 | 169 | 22 | 53 |
| 166 | 170 | 21 | 90 |
| 167 | 171 | 20 | 73 |
| 168 | 172 | 20 | 52 |
| 169 | 173 | 19 | 98 |
| 170 | 174 | 20 | 56 |
| 171 | 175 | 19 | 84 |
| 172 | 176 | 20 | 77 |
| 173 | 177 | 18 | 97 |
| 174 | 178 | 21 | 76 |
| 175 | 179 | 19 | 94 |
| 176 | 180 | 20 | 76 |
| 177 | 181 | 19 | 56 |
| 178 | 182 | 21 | 58 |
| 179 | 183 | 22 | 59 |
| 180 | 184 | 21 | 87 |
| 181 | 185 | 20 | 50 |
| 182 | 186 | 20 | 85 |
| 183 | 187 | 19 | 83 |
| 184 | 188 | 20 | 40 |
| 185 | 189 | 19 | 89 |
| 186 | 190 | 19 | 75 |
| 187 | 191 | 20 | 77 |
| 188 | 192 | 18 | 44 |
| 189 | 193 | 21 | 75 |
| 190 | 194 | 19 | 74 |
| 191 | 195 | 20 | 43 |
| 192 | 196 | 19 | 74 |
| 193 | 197 | 21 | 74 |
| 194 | 198 | 22 | 53 |
| 195 | 199 | 21 | 87 |
| 196 | 200 | 20 | 75 |
| 197 | 201 | 20 | 42 |
| 198 | 202 | 19 | 80 |
| 199 | 203 | 20 | 46 |
| 200 | 204 | 19 | 91 |
| 201 | 205 | 20 | 78 |
| 202 | 206 | 18 | 80 |
| 203 | 207 | 21 | 79 |
| 204 | 208 | 19 | 91 |
| 205 | 209 | 20 | 70 |
| 206 | 210 | 19 | 59 |
| 207 | 211 | 21 | 58 |
| 208 | 212 | 22 | 56 |
| 209 | 213 | 21 | 92 |
| 210 | 214 | 20 | 54 |
| 211 | 215 | 20 | 93 |
| 212 | 216 | 19 | 95 |

|     |     |    |    |
|-----|-----|----|----|
| 213 | 217 | 20 | 53 |
| 214 | 218 | 19 | 96 |
| 215 | 219 | 19 | 75 |
| 216 | 220 | 20 | 75 |
| 217 | 221 | 18 | 52 |
| 218 | 222 | 21 | 74 |
| 219 | 223 | 19 | 79 |
| 220 | 224 | 20 | 50 |
| 221 | 225 | 19 | 72 |
| 222 | 226 | 21 | 71 |
| 223 | 227 | 22 | 51 |
| 224 | 228 | 21 | 99 |
| 225 | 229 | 20 | 70 |
| 226 | 230 | 20 | 55 |
| 227 | 231 | 19 | 97 |
| 228 | 232 | 20 | 59 |

*##Q2.k) Student\_IDs which are present in Students(dfs) but not in Marks(dfm)*

```
missing_ids = dfs[~dfs["Student_id"].isin(dfm["Student_id"])]
missing_ids
```

|    | Student_id | Age | Gender | Grade     | Employed |
|----|------------|-----|--------|-----------|----------|
| 8  | 9          | 22  | Female | 3rd Class | yes      |
| 9  | 10         | 21  | Male   | 1st Class | no       |
| 48 | 49         | 19  | Male   | 2nd Class | no       |

*## Q3. Concept of missing values*

Missing Values in a Pandas Dataframe - results when no data stored for a particular variable

Missing values are generally represented as NaN (Not a Number)

Missing values can be identified using: isna() or isnull() or info()

The isna() function in Pandas is used to detect missing (NaN) values in a DataFrame or Series.

It returns a DataFrame/Series of the same shape with Boolean values (True/False)

DataFrame.isna() / Series.isna()

print(df.isna())

isnull() and isna() are identical in pandas and can be used interchangeably.

print(df.isnull().sum()) - Returns the count of missing values per column

print(df.info()) - Shows the number of non-null values in each column

Handling Missing Values:

-----

There are different techniques for handling missing values based on the context and dataset:

a) Deleting Missing Values:

```

    Drop rows with missing values --> df_dropped_rows = df.dropna()
    Drop columns with missing values --> df_dropped_cols =
df.dropna(axis=1)

b) Imputing Missing Values:
    - Fill with a constant value --> df_filled = df.fillna(0)
    - Fill with column mean/median/mode -->
df['A'].fillna(df['A'].mean(), inplace=True) # Mean Imputation

df['B'].fillna(df['B'].median(), inplace=True) # Median Imputation

df['C'].fillna(df['C'].mode()[0], inplace=True) # Mode Imputation

    - Forward Fill (ffill) or Backward Fill (bfill) -->
df.fillna(method='ffill', inplace=True) # Fill using previous value

    df.fillna(method='bfill', inplace=True) # Fill using next value
    - Using Interpolation --> df.interpolate(method='linear',
inplace=False, limit_direction='forward')
    to estimate and fill missing values (NaN) in a DataFrame or Series
    by predicting the missing values based on surrounding data points
    method: Specifies the interpolation method (e.g., linear,
    polynomial, spline, etc.)
    limit_direction: Can be 'forward', 'backward', or 'both' to
    specify the direction of interpolation
    inplace: If True, modifies the DataFrame in place

Example:
import pandas as pd

# Creating a sample DataFrame
df = pd.DataFrame({'A': [1, 2, None, 4, 5],
                   'B': [None, 3, 4, None, 6],
                   'C': [1, None, None, 4, 5]})

# Identifying missing values
print("Missing Values:\n", df.isnull().sum())

# Handling missing values
df['A'].fillna(df['A'].mean(), inplace=True) # Imputing mean
df['B'].fillna(method='bfill', inplace=True) # Backward fill
df['C'].fillna(df['C'].mode()[0], inplace=True) # Mode
print("\nUpdated DataFrame:\n", df)

```